Abstract

The project is devoted to the design and optimization of an AI-supported ESG (Environmental, Social, and Governance) reporting tool that are used to provide sustainability disclosures to achieve the EFRAG VSME standard that is implemented by smaller and medium-sized companies. The conventional processes associated to ESG reporting are time-consuming, inaccurate, and incapable of real-time characteristics thereby restricting the extent of regulatory compliance and utility of decisions. The research aims to overcome these pitfalls by using artificial intelligence tools including machine learning, and natural language processing to automate data collection, check the data, measure environmental and social metrics and produce precise ESG reports. The project combines the data provided in financial records, demographics of the workforce, energy and water management, waste management, and governing practices. The major data point out the following: 120 billion MWh total energy consumption, GHG intensity of 113.7 tCO2e per one million euro turnover, recycling non-hazardous waste of 92.7 percent. Its workforce is constituted of 50 employees, 88 percent of workforce in permanent contracts, and 4.5 percent gender pay gap and 12 percent turnover rate.

The current research aims at exploring how an Artificial Intelligence (AI) enabled Environmental, Social and Governance (ESG) reporting system can be designed and applied specifically to Very Small and Medium Enterprises (VSMEs). The study is relevant given that rising regulatory pressure and stakeholder pressure to disclose sustainability transparently have led to the need to examine how artificial intelligence can be used to improve data collection, analysis, and compliance with ESG requirements in resource-strapped organisations. Based on the European Financial Reporting Advisory Group (EFRAG) VSME framework, the study adopts a mixed-methods research design, which includes system testing, quantitative benchmarking, and qualitative assessment. As a result, the findings point out that the AI tool greatly streamlines governance oversight by detecting reporting inconsistencies and ensuring concordance with ESG regulations. The study arrives at the conclusion that AI-driven ESG reporting proves to be a game-changing measure in helping VSMEs not only meet regulatory requirements but also manage sustainability as a whole. Future research ought to focus on globalization, autonomous learning algorithms, and inter-industrial applicability to maximise the overall long-term impact of the system.

Table of Contents

Chapter 1: Introduction 1

1.1 Research Background 1

1.2 Problem Definition 2

1.3 Aim and Objectives 3

1.4 Research Question 3

1.5 Significance of the Research 4

1.6 Thesis Structure 5

Chapter 2: Literature Review 6

2.1 Introduction 6

2.2 Evolution of ESG Reporting Standards and Frameworks 6

2.3 Challenges Faced by VSMEs in ESG Compliance 9

2.4 Role of Artificial Intelligence in Sustainability and Compliance 10

2.5 Automated Data Collection and ESG Metric Extraction 12

2.6 Natural Language Processing for ESG Report Generation 15

2.7 Materiality Assessment through AI and Machine Learning 16

2.8 Comparative Analysis of Existing ESG Reporting Platforms 18

2.9 Literature Gap 21

Chapter 3: Research Methodology 23

3.1 Research Philosophy 23

3.2 Research Approach 24

3.3 Research Design 25

3.4 Data Analysis and Collection Methods 28

3.5 Ethical Consideration 29

3.6 Summary 30

Chapter 4: Research Findings 31
4.1 Introduction 31
4.2 Result Analysis 31
4.3 Key Findings 48
4.4 Critical Analysis 51
4.5 Discussion 53
4.6 Summary 55
Chapter 5: Conclusion 57
5.1 Linkage to Objectives 57
5.2 Summary 57
5.3 Reflection 58
5.4 Research limitation 59
5.5 Recommendations 59
5.6 Future work 61
Reference List 62
Appendix 73
Declaration of Authenticity 74

List of Figures

Figure 1.6.1: Structure of Thesis 5
Figure 2.2.1: Evolution of ESG Reporting Standards and Frameworks 7
Figure 2.3.1: Challenges Faced by VSMEs in ESG Compliance 10
Figure 2.5.1: Automated Data Collection and ESG Metric Extraction 13
Figure 2.8.1: Comparative Analysis of Existing ESG Reporting Platforms 19
Figure 3.3.1: AI-Assisted ESG Reporting Design Flowchart 26
Figure 4.2.1: Data Sources for VSME Reporting 32
Figure 4.2.2: Data Collection Window 32
Figure 4.2.3: Data Collection View with Chapters 33
Figure 4.2.4: Basic Sensitive Information for The ESG Financial Report 34
Figure 4.2.5: Balance Sheet and Turnover in Euro 34
Figure 4.2.6: Circular Economy Principles for Waste Management and Resource Usage 35
Figure 4.2.7: Corruption Analysis for ESG Reporting 36
Figure 4.2.8: Future Sustainability Initiatives and Monitoring Targets 37
Figure 4.2.9: Total Energy Consumption 37
Figure 4.2.10: GHG Emissions and Intensities 38
Figure 4.2.11: Details on The Pollution of Air, Water, and Soil 39
Figure 4.2.12: Specification of Biodiverse Sensitive Area 40
Figure 4.2.13: Total Use of Land to Maintain Biodiversity 41
Figure 4.2.14: Sealed and Nature-Oriented Area 41
Figure 4.2.15: Water Withdrawal Specifications 42
Figure 4.2.16: Non-Hazardous Waste Breakdown 43
Figure 4.2.17: Hazardous Waste Breakdown 43
Figure 4.2.18: Contractual Workforce 44
Figure 4.2.19: Gender-wise Workforce 45
Figure 4.2.20: Workforce Breakdown in Terms of The Country and Employee Turnover Rate . 46
Figure 4.2.21: Health and Safety for The Workforce 46
Figure 4.2.22: Remuneration and Wage Gap 47
Figure 4.2.23: Average Annual Training Hours 48

List of Tables

Table 2.2.1: Overview of Key ESG Reporting Frameworks and Their Relevance 9
Table 2.4.1: Applications of Artificial Intelligence in ESG Sustainability and..... 12
Table 2.5.1: Technologies for Automated ESG Data Collection and Extraction 15
Table 2.7.1: AI-Driven Materiality Assessment Features in ESG Reporting..... 18
Table 2.8.1: Comparative Analysis of Selected ESG Reporting Platforms..... 21
Table 3.3.1: Summary of Research Design 27
Table 4.5.1: AI-Based ESG Reporting – Key Discussion Points 55

Chapter 1: Introduction

1.1 Research Background

"Environmental, Social, and Governance" reporting is becoming of great concern to companies, investors, and stakeholders to understand sustainability and the ethical impact of an organization. ESG reporting provides details of how companies manage risks and opportunities of environmental stewardship, social responsibility, and governance. The growing pressure by regulators, consumers, and investors to have greater disclosure has led to the adoption of a myriad of ESG reporting frameworks and tools among organizations to measure, report, and improve their sustainability performance. Traditional ESG reporting is carried out manually, involving time-consuming analysis, as well as centrally-dispersed systems (Wang, 2025). These methods are relatively erroneous and inefficient. The traditional "ones do not handle such large volumes of data with varying sources well, hence giving reports at the wrong time or with the wrong information. With businesses needing to keep pace with the evolving forms and requirements of regulation and demonstrate their own environmental and social performance, there seems to be an urgent necessity to have ESG reporting on a larger and more efficient scale.

Artificial Intelligence (AI) is new, which offers a way to transform ESG reporting with automation of data collection, processing, and analysis. With ML, NLP, and predictive analytics, AI tools can retrofit ESG reporting tenfold by providing real-time data and highlighting patterns and issuing accurate reports with a minimum human involvement (Foziljonov et al. 2025). Pattern or risk detection can also be automated in AI, which gives a more forward-looking and powerful response to ESG performance management. This transformation to AI-based reporting achieves a reduction in errors, a faster production of reports, and delivers actionable knowledge to the stakeholders.

The graph illustrates how "Artificial Intelligence (AI)" fits in the enhancement of ESG "(Environmental, Social, and Governance) reporting". The AI is used to make this process easier by automating such important actions as Information Mining, in which relevant ESG information is extracted across various sources in a brief time frame with high accuracy. Actionable Intelligence is used to provide relevant insight in order to assist companies in making well-informed sustainability decisions. AI also facilitates Information Extraction from Images and Tables, thus vital information from intricate documents gets processed effectively. With Assisted Scoping, AI determines appropriate data, narrowing the scope of reports. Entity Recognition is used to study past data, providing an insight into historic ESG trends, while the Prediction of

Compliance Probability enables organizations to determine their compliance with rules (De Villiers et al. 2024). The Interactive Q&A Engine for Intelligent Searching enables users to ask and receive ESG-related information, increasing data accessibility. This AI integration heavily optimizes and streamlines ESG reporting for organizations.

VSME reporting is a new paradigm that embeds AI to facilitate ESG reporting. VSME enhances decisions and promotes sustainable and standard (norm-compliant) ESG reports production and rating in real-time and automatically. However, developing an AI-based VSME-based ESG reporting tool has to overcome technical challenges related to data aggregation, model selection, and feature creation. This PhD constructs an AI-enabled system of ESG reporting with a center-stage VSME reporting model to determine major design elements and AI implementation to create a comfortable, easy, and legally compliant ESG reporting machinery.

1.2 Problem Definition

The unavailability of effective, accurate, and automated ESG reporting tools is the key issue addressed in this thesis. Traditional processes of ESG reporting are typically time-consuming and manual in nature and rely on steps of data collection and analysis, which remain prone to delays and errors. The need to have more efficient solutions increases with growing demands to see companies reporting on ESG topics in a transparent and precise manner (Hsu et al. 2025). It makes meeting the regulatory requirements and providing valuable information to the stakeholders quite difficult.

Lack of standardized and automated tools and frameworks is considered one of the key obstacles in the improvement of ESG reporting. Although there exist various reporting standards such as the “Global Reporting Initiative (GRI) or Sustainability Accounting Standards Board (SASB)”, it more often than not lacks the capability of integration of the data on a real-time basis or automation. The ESG data requires companies to compile information manually, and this results in inefficiency and an increased possibility of making errors. Moreover, the lack of automation limits the possibility of monitoring ESG performance in real-time..

Thereafter, artificial intelligence-based applications deal with these barriers by automating data monitoring, research, and ESG report generation. There remain numerous blank spaces in how to fashion such improvements, especially how AI is inserted into such systems as VSME reporting (Mitra & Maity, 2025). The specific question is what skills are the most essential to effective ESG reporting tools, what type of data sources are incorporated, and how AI models are transformed to

perform beneficial, actionable information. This thesis aims to look into these challenges and develop an ESG reporting solution with an AI-enabled approach of the VSME framework.

1.3 Aim and Objectives

Aim

This thesis aims to create an ESG reporting tool based on AI principles, based on the VSME reporting model. It is given priority is automate data collection, realize real-time analysis, and generate insights that are applied to improve ESG performance reporting in organizations.

Objectives

- ***To Award Key Features Required of an AI-Assisted ESG Reporting Tool*:** Using the assistance of AI, determine the bare requirements and elements within a skilled ESG reporting system.
- ***To Implement a Structure of Inclusion of VSME Reporting in the Tool*:** Create a framework of incorporation of VSME reporting methodology in the ESG reporting tool to fit in easily.
- ***To Determine the Effect of AI Integration on ESG Reporting Precision*:** Analyze how AI integration enhances the precision, reliability, and accuracy of ESG reports.
- ***To Create a User Interface for Streamlined ESG Report Creation*:** Create an easy-to-use user interface that allows for the simple creation and visualization of ESG reports.

1.4 Research Question

1. What AI technologies and approaches can streamline ESG reporting through automation, error minimization, and accuracy?
2. Which is the most important feature, such as automation, real-time data integration, or user interface design, that is required for an effective ESG reporting tool?
3. How does the VSME reporting system enable automation of ESG reporting and improve reporting efficiency?
4. What are the technical, data-related, and ethical challenges in developing AI models for analyzing and reporting ESG data?
5. How can AI-enabled tools enhance ESG compliance and actionable insights for informed decision-making in organizations?

1.5 Significance of the Research

The importance of this study is that it has the ability to revolutionize the manner in which Environmental, Social, and Governance (ESG) reporting is carried out, especially through the implementation of Artificial Intelligence (AI) into VSME reporting. As sustainability takes center stage as an increasingly important issue to the attention of companies, governments, and investors, there is increasing pressure on organizations to disclose transparent, accurate, and timely ESG data (Cao et al. 2024). The legacy strategies to ESG reporting, which are typically manual and labour-intensive, are lengthy, inaccurate, and lack the extent of disclosure that is necessary to comply with regulations and the expectations of interested parties.

Since AI is being introduced into ESG reporting, this study presents a potential solution to such problems. When human-powered mechanisms are substituted by the power of AI, not only does the possibility of human factors that contribute to the occurrence of errors decrease, however large reserves of time are saved, as well (Efe, 2023). Artificial Intelligence is capable of dealing with large volumes of data in real-time, which enables organizations to have timely information on their ESG performance. Such real-time reporting abilities enable more rapid and more informed decision-making and responsiveness to changes in ESG metrics or changes in the regulatory environment.

Another innovation of the ESG reporting procedure is the implementation of VSME (Virtual Sustainable Management and Evaluation) reporting. VSME is a system whose main concern is the sustainable management methods provided through the virtual and automated method, and that is exactly where the AI-enabled reporting strives to go. With VSME turning ESG reporting into an automated process, firms are able to ensure that not only are their reports internationally compliant, but also tailored to suit their particular sustainability goals (Tong et al. 2022). This helps organizations to track their progress with greater accuracy and make decisions that are prone to information that supports their long-term sustainability principles.

This study has a multidimensional contribution and also enhances efficiency in ESG reporting. It enhances accuracy as automation gives results and reports that are trustworthy. AI can also help organizations achieve compliance with the regulations by providing real-time and accurate analytics about their current ESG progress. AI allows better decision-making as it provides recommendations that can then be used by businesses to make well-grounded decisions on their sustainability projects (Boedijanto & Delina, 2024). The paper contributes to the innovation of the

software involved in ESG reporting. There is an emerging interest in the automation of the system and increased sustainability performance in organizations. The research provides a positive new step toward ESG reporting being easier, more precise, and more understandable to act upon, and it helps businesses and stakeholders everywhere.

1.6 Thesis Structure

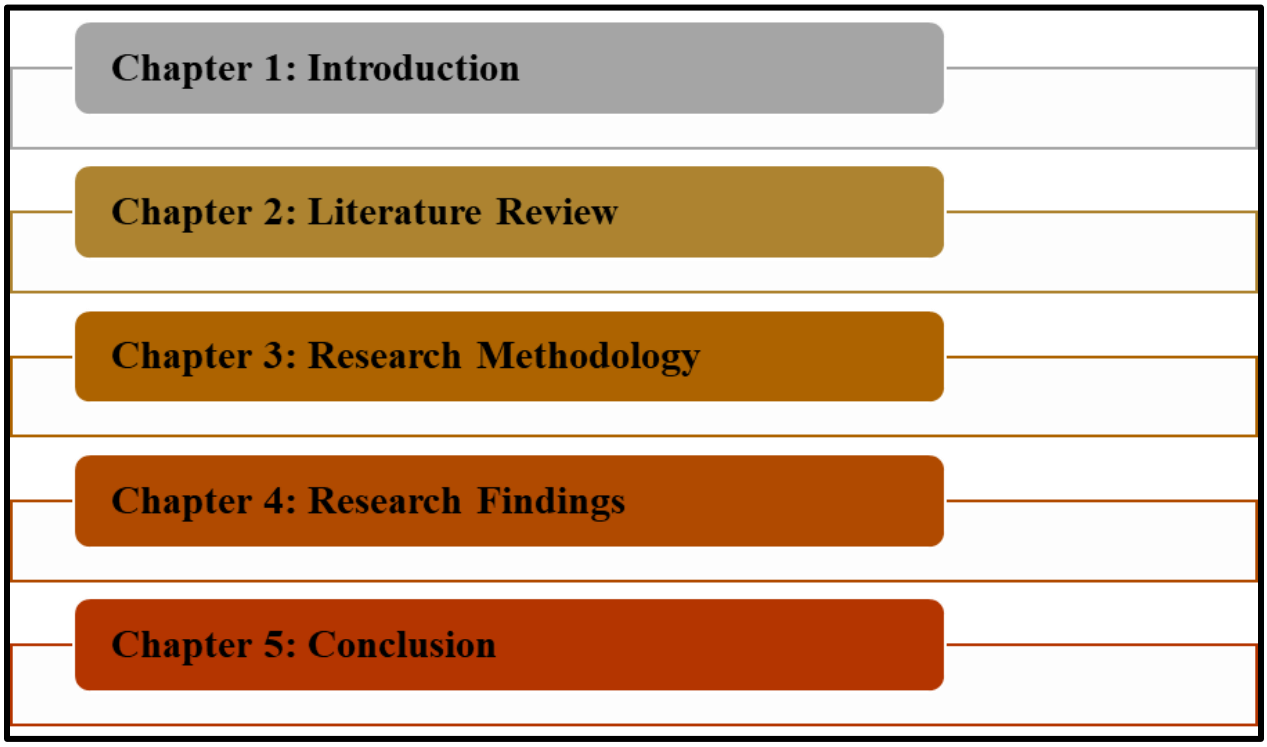


Figure 1.6.1: Structure of Thesis

Chapter 2: Literature Review

2.1 Introduction

The integration of Environmental, Social, and Governance (ESG) factors into business strategy today is the central mandate for businesses seeking to meet regulatory requirements, stakeholder demands, and sustainability aspirations. Large enterprises, in most circumstances, have basic infrastructure to keep tasks involved in complex ESG reporting requirements tractable, while very small and medium-sized enterprises (SMEs) are hindered by their limited resource base and administrative muscle. The ESRS VSME reporting model tries to minimize the burden through simplified standards, but regulation is a mammoth hurdle without required technological support. Artificial Intelligence (AI) is gradually turning into a game-changer in the automation of ESG compliance, the accuracy of disclosures, and the facilitation of data-informed sustainability behaviours. To automate the work on the reports and to deprive them of human interference, as well as generate real-time ESG insights, among the AI applications, which are data extraction, natural language processing (NLP), and predictive analytics, one can consider.

2.2 Evolution of ESG Reporting Standards and Frameworks

The development of Environmental, Social and Governance (ESG) reporting standards shows an increase in demand for formalised and transparent disclosures of corporate sustainability performance. There is an increasing trend towards standardization of the ESG disclosure that has been initially fragmented and on a voluntary basis, occasioned by global standards formulated in the best interest of comparability, credibility, and stakeholder value. One of the most prolonged and applied ESG guidelines is the Global Reporting Initiative (GRI) (Gee, 2025). Designed to make sustainability reporting uniform geographically and industry-wise, the GRI framework is based on qualitative and quantitative reporting on sustainability impacts. It focuses on stakeholder identification, materiality, and the holistic sweep of corporate impact on society and the environment. Its modularity and flexibility have subjected it to widespread adoption, particularly by multinational companies. The Sustainability Accounting Standards Board (SASB) began with an industry-based framework. Its industry-specific design for detecting financially significant sustainability concerns is SASB's. SASB's emphasis on materiality from the investor perspective is consistent with the expanding application of ESG metrics to make investment decisions.

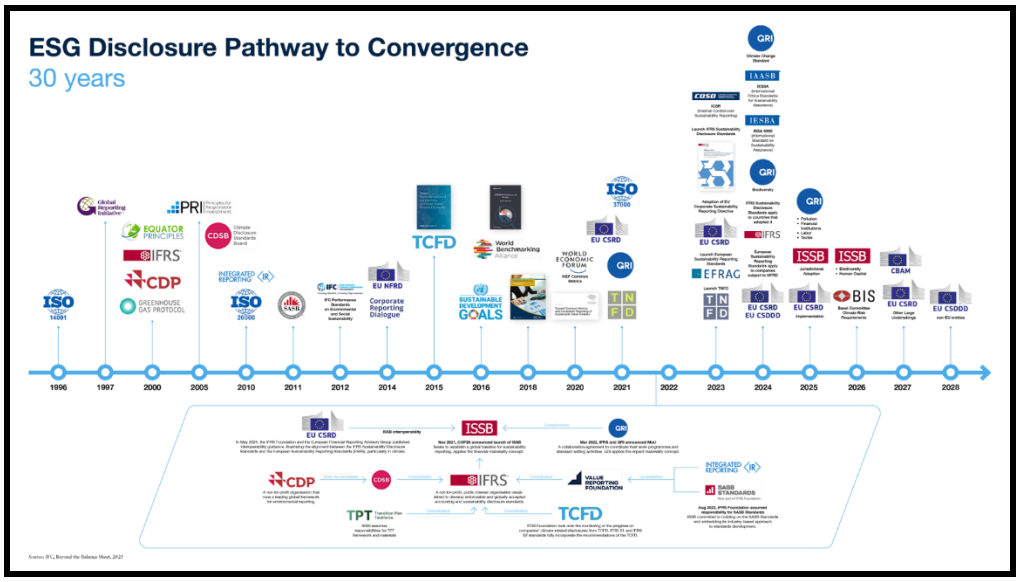


Figure 2.2.1: Evolution of ESG Reporting Standards and Frameworks

(Source: Lehtimäki, 2024)

The Task Force report on Climate-Related Financial Disclosures (TCFD) has been geared towards increasing the credibility of climate-related opportunities and risks within the financial system. The TCFD guidance revolves around four thematic streams: governance, strategy, risk management, and metrics and targets. The TCFD process is intended to promote scenario analysis and forward-looking disclosure; hence, it is mandatory while dealing with climate risk and transition planning. The European Union has also strengthened the regulatory framework through the Corporate Sustainability Reporting Directive (CSRD). It supersedes the earlier Non-Financial Reporting Directive (NFRD), and the CSRD enlarges more than twice the ESG reporting obligations across the EU. It mandates the utilization of European Sustainability Reporting Standards (ESRS), which are prepared by EFRAG, to make it standardized and guaranteed. The ESRS lays down detailed stipulations on environmental, social, and governance matters that increase comparability and uniformity of ESG information. In acknowledgement of the burden of compliance by small entities, the ESRS framework provides reduced reporting templates for small and medium-sized enterprises (SMEs) and micro-enterprises (Muck & Schmidl, 2024). Regulations like ESRS LSME (for big SMEs) and VSME (for very small entities) aim to minimize complexity while underlying disclosure principles are retained.

Framework	Developed By	Focus Area	Key Characteristics	Relevance to VSMEs
GRI (Global Reporting Initiative)	Global Sustainability Standards Board (GSSSB)	Broad ESG disclosure	Stakeholder-focused, global applicability, modular structure	Offers flexibility; complex for VSMEs without simplification
SASB (Sustainability Accounting Standards Board)	Value Reporting Foundation (now under IFRS Foundation)	Industry-specific materiality	Sector-specific ESG issues, financially material metrics, and investor-oriented	Detailed materiality maps are useful for streamlined SME reporting
TCFD (Task Force on Climate-Related Financial Disclosures)	Financial Stability Board	Climate risk disclosure	Forward-looking metrics, scenario analysis, and risk management integration	High relevance for firms in climate-sensitive sectors; technical for VSMEs
CSRD (Corporate Sustainability Reporting Directive)	European Commission	Regulatory ESG disclosure	Mandatory for EU companies, integrates double materiality, and assurance is required	Applies to larger SMEs; phase-in for smaller entities through tailored standards

ESRS (European Sustainability Reporting Standards)	EFRAG (European Financial Reporting Advisory Group)	EU-wide ESG standardisation	Detailed topic-specific standards align with CARD, including simplifications	ESRS LSME and ESRS VSME are under development to reduce complexity for small firms

Table 2.2.1: Overview of Key ESG Reporting Frameworks and Their Relevance

2.3 Challenges Faced by VSMEs in ESG Compliance

Very Small and Medium-Sized Enterprises (VSMEs) also possess unique structural and functional issues that hinder productive participation in Environmental, Social, and Governance (ESG) compliance. Despite increased regulatory pressure for ESG disclosure systems, effective use within VSMEs is hindered by systemic obstacles. Human and financial resources are only a few of the key obstacles since these lack adequate support (Tian et al. 2024). Compared to large businesses that have the luxury of establishing specialist budgets and ESG reporting and specialist strategy teams, VSMEs adopt lean organisational structures. Financial investment in VSMEs tends to focus on short-term operational needs at the cost of long-term sustainability initiatives. Consequently, ESG initiatives appear as a cost element with little short-term payback returns and with a concomitant low priority. The second major hindrance is the internal absence of ESG knowledge. Qualified sustainability officers, attorneys, or ESG compliance specialists familiar with reading and implementing complex reporting requirements are never employed by VSMEs. The absence of skills introduces uncertainty about what ESG data is appropriate, how to obtain necessary information, and how to link disclosures to evolving regulatory needs.

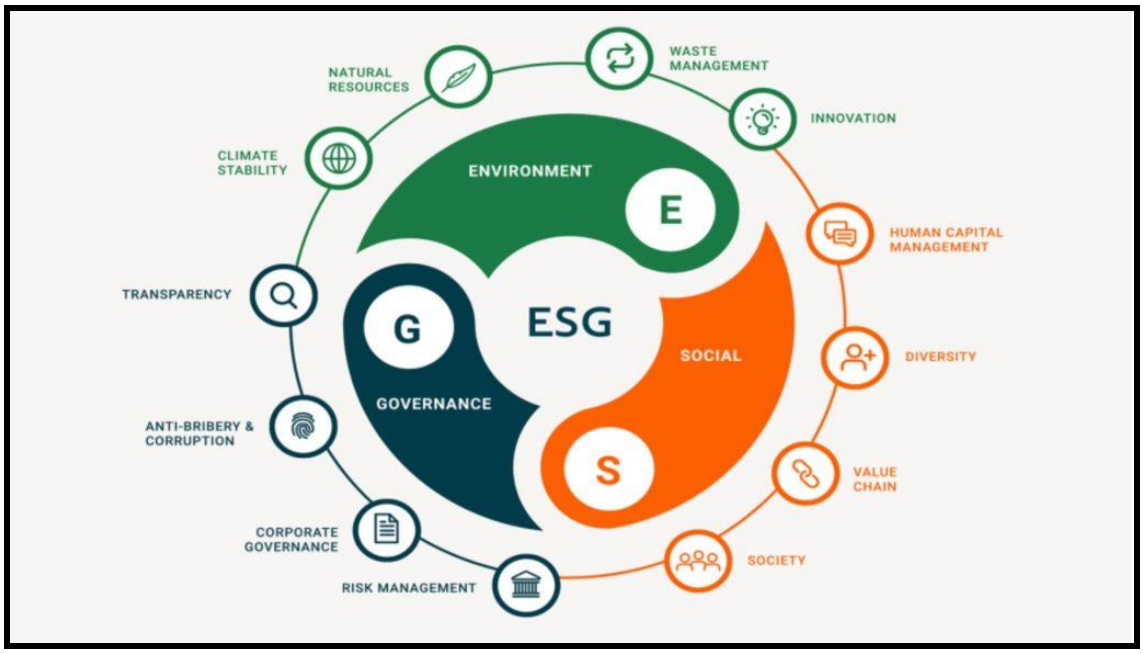


Figure 2.3.1: Challenges Faced by VSMEs in ESG Compliance

(Source: Sharma et al. 2024)

The situation is aggravated by fragmentation and the simplicity of the ESG regulatory environment. The wide range of standards available is complex; their definitions, coverage, and forms of reporting vary: there are global approaches, including TCFD and GRI, and regional ones, such as the EU Corporate Sustainability Reporting Directive (CSRD). In the case of VSMEs, such diversity causes confusion and a surplus of unnecessary administrative burden. The changeover between interdependent necessities, in the absence of professional help, is always characterized by imperfect or non-conformant reporting. The tools need to be designed in such a way that, in case of manual manipulation, the technicality level has been minimal, and there is an avenue that even without expert knowledge of ESG users can follow. Among some of the features in reducing the compliance barrier of ESG is the pre-defined templates of VSME disclosures, automated data collection, and regulatory heatcheck in the tools (Wijaya et al. 2025). The tools ought also to enable scalability in adoption, and VSMEs ought to be in a position to add tiers of depth and quality to their sustainability reporting as these grow in organisational maturity.

2.4 Role of Artificial Intelligence in Sustainability and Compliance

AI has served to be transformational in both reinforcement of sustainability objectives and regulation stipulations through the application of Environmental, Social, and Governance (ESG) practices. In recent years, the potential opportunities of AI-solutioned solutions to enhance the

accuracy, efficiency, and responsiveness in the ESG reporting systems have demonstrated an enormous potential, especially within complex and changing rules-based regimes (Xianzi et al. 2025). On the list of most common AI uses in ESG is risk analysis. Advanced machine-learning models can process massive amounts of structured and unstructured data to detect emerging risks of sustainability issues in the supply chains and business operations, and geopolitics. The technologies allow the firms to monitor environmental indicators, indicators in labour policies, and indicators of governance in real-time and help to act better in the short-term reaction as well as to plan in the long-term perspective. Green has been hiding may be detected with the help of AI, and the validity of sustainability claims can be proved. Natural Language Processing can be applied to examine public disclosures, sustainability reports, and third-party reports for inconsistencies or hyperbole.

Predictive analytics models leverage the historical data and external benchmarks to predict future sustainability performance, e.g., carbon emission trajectory or diversity ratio. The data supports strategic decision-making and helps organizations get in tune with global sustainability targets, e.g., the UN SDGs. Scalability, adaptability, and real-time processing are benefits of AI adoption into ESG reporting. AI solutions can encompass the entire spectrum of worldwide operations, such as adherence to geographic rules of regulation and reporting formats. Real-time processing and analysis of big data ensure punctuality in updates and adherence, reducing the burden of manual reports and operational inefficiencies. While such solutions are trendy in multinational corporations, the same technologies and strategies are becoming compatible with Very Small and Medium Enterprises (VSMEs). Affordable AI platforms, open-source technology, and cloud facilities allow VSMEs to implement AI-powered ESG solutions with little initial capital investment (Li et al. 2025). Simplified interfaces and automated functions are being made available to meet the special needs of smaller organisations, which may lack specialist ESG personnel, thereby making state-of-the-art compliance tools more widely accessible and stimulating greater participation in sustainable business operations.

AI Application	**Function**	**Benefit**	**Relevance for VSMEs**

Risk Assessment	Identifies sustainability-related risks using data analytics and forecasting	Early detection of ESG risks	Enables proactive management with limited resources
Green has been hiding Detection	Uses NLP to analyze reports for inconsistent or misleading sustainability claims	Ensures credibility and transparency	Protects reputation in compliance-sensitive markets
ESG Performance Prediction	Forecasts future ESG outcomes using historical and benchmark data	Supports long-term planning and benchmarking	Assists in setting realistic sustainability targets
Real-Time Data Processing	Continuously ingests and analyses ESG data streams	Immediate insights for decision-making	Reduces manual effort and ensures timely reporting
Scalability and Adaptability	AI models can be trained and adjusted for various regulatory standards	Uniformity in multi-jurisdictional reporting	Offers flexible compliance for diverse stakeholders
Automation of Reporting Tasks	Automates the generation of ESG narratives and data aggregation	Improves accuracy, saves time	Essential for small teams with limited expertise

Table 2.4.1: Applications of Artificial Intelligence in ESG Sustainability and

2.5 Automated Data Collection and ESG Metric Extraction

Artificial data gathering is fairly common in modern ESG reporting, particularly among resource-constrained VSMEs. Utilizing sophisticated technologies such as Application Programming Interfaces (APIs), robotic process automation (RPA), and smart sensors facilitates seamless extraction and tracking of significant ESG indicators (Ilieva et al. 2025). These technologies support real-time tracking of such information as energy consumption, greenhouse emissions, and employee diversity metrics, which are of the greatest importance to stay compliant with evolving ESG demands. APIs enable real-time exchange of ESG-critical information among internal and

external systems. Utility providers' APIs, for instance, can pull in carbon usage or electricity usage data directly, while HR systems can supply demographic profiles for diversity and inclusion analyses. RPA goes even further to transport or automate repetitive reconciliations of information between systems to minimize human errors and enhance uniformity of information on different categories of ESG. Used concurrently, these blocks of building allow organizations to collect ESG data with as little human interference as possible. Another contributor of ESG information is Web scraping and Optical Character Recognition (OCR), which is just as significant.

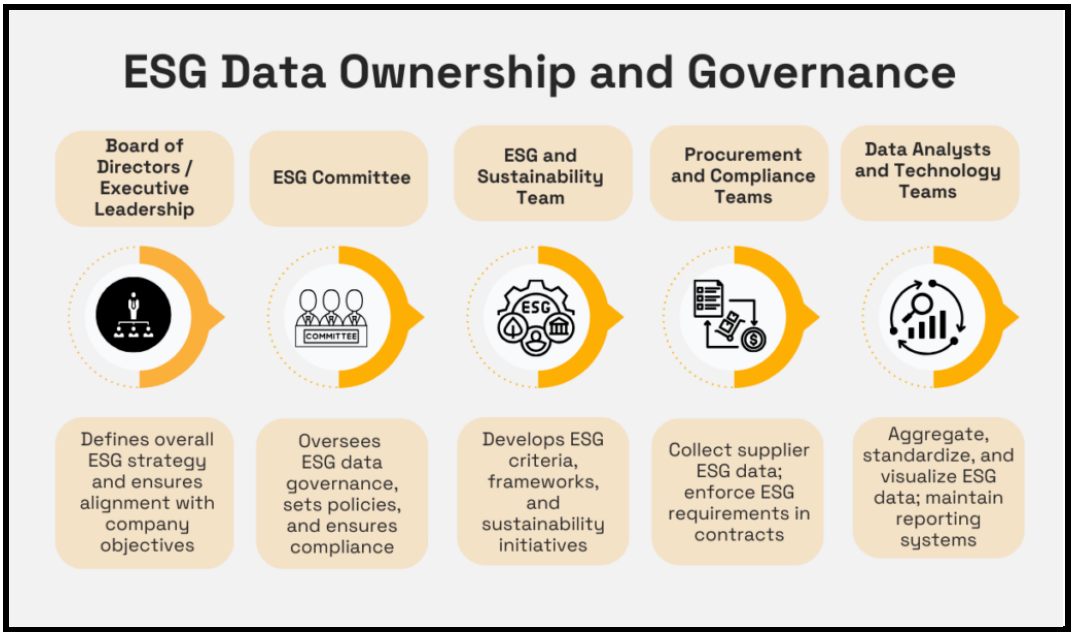


Figure 2.5.1: Automated Data Collection and ESG Metric Extraction

(Source: Attia, 2024)

OCR can be used to scan paper-based reports or print out ESG reports or supplier statements to make the unstructured text analyzable and machine-readable. In the meantime, web scraping methodologies can retrieve ESG-related disclosures in publicly available websites like sustainability portals, reports filed with governments, or even websites of competitors. These functionalities are especially useful to generate contextual data or benchmarking of ESG performances without the necessity to conduct huge amounts of manual research. The possibility of ESG reporting being automated is one of the key areas, as field AI models induce the indicators of structured and unstructured data. Machine learning approaches that have been trained on the basis of ESG taxonomies would be capable of ensuring that such material would be sorted out into environment-, social-, or governance-bins and accurately mark out materiality or non-compliance issues (Luccioni et al. 2025). The processing with NLP can also derive the sentiment or keyword

frequency, or policy compliance out of the textual material, thus, easily automates the meaning of the ESG in context.

Technology	Function	Example ESG Metrics Extracted	Relevance to VSMEs
APIs (Application Programming Interfaces)	Enables automated data flow between systems (e.g., utility companies, HR platforms)	Carbon footprint, electricity usage, workforce diversity	Reduces manual data input by integrating with existing systems
Robotic Process Automation (RPA)	Automates repetitive data entry and report aggregation tasks	Data compilation from spreadsheets and report formatting	Minimizes time and human error in reporting workflows
Sensors and IoT Devices	Captures real-time environmental performance data	Water usage, energy consumption, and temperature control	Provides accurate, real-time tracking of operational sustainability
Web Scraping	Extracts ESG data from public online sources	Competitor ESG disclosures, regulatory updates	Supports benchmarking and regulatory alignment
Optical Character Recognition (OCR)	Converts printed or scanned documents into digital text	Supplier certificates, compliance letters	Enables digital tracking of offline ESG documentation
Enterprise System Integration (e.g., ERP)	Pulls structured ESG-relevant data from core business applications	Resource usage, procurement practices, and financial KPIs	Leverages existing internal databases for ESG metrics

AI-Powered Data Classification	Classifies and extracts ESG indicators from structured and unstructured data	Sustainability narratives, governance disclosures	Enhances accuracy and interpretation in reporting
Natural Language Processing (NLP)	Analyzes text data for ESG-related terms, sentiment, and materiality	Policy statements, stakeholder feedback, and audit reports	Automates qualitative analysis for ESG report narratives

Table 2.5.1: Technologies for Automated ESG Data Collection and Extraction

2.6 Natural Language Processing for ESG Report Generation

Traditional ESG reporting often involves an enormous amount of manual work to review datasets, extract relevant indicators, and convert them into formal disclosures (Strang & Vajjhala, 2024). NLP surmounts these limitations by searching in unstructured text, discovering ESG-related entities, and creating well-meaning summaries out of material sustainability concerns. Latest advances in large language models (LLMs) like GPT-based architecture have also enhanced the application of NLP in producing ESG reports. Such models not only possess the capability to produce human-like text but also contextual knowledge of ESG terms such as Scope 1, 2, and 3 emissions, gender diversity metrics, or board independence percentages. By training these models on ESG templates, regulation reports, and previous reports, it is possible to make them produce full disclosure paragraphs of climate risk stories, environmental impact analyses, and social governance policies of high language quality and domain relevance.

LLMs have also been shown to be effective in assisting in producing climate disclosures, which could be required to be made under specific reporting requirements such as the Task Force on Climate-related Financial Disclosures (TCFD) or Corporate Sustainability Reporting Directive (CSRD). The tools can be programmed to automatically produce structured responses to reporting templates in the form of numbers and qualitative data combined into a consistent tone and voice. Also, NLP models can adjust the lexicon and register of language in real-time for application among diverse stakeholder groups, say investors, regulators, or internal management. To Very Small and Medium Enterprises (VSMEs), the benefits of such NLP-powered functionalities are particularly significant. These business entities generally lack in-house sustainability capability

and face acute resource constraints. NLP-based automation enables them to produce professionally written, standardized ESG reports without the need for permanent reporting staff (Twaha et al. 2025). Furthermore, by reducing the dependency on manual interpretation and editing, these systems enhance reporting consistency, turnaround time, as well as minimize the possibility of human mistakes or non-compliance.

2.7 Materiality Assessment through AI and Machine Learning

Determination of materiality is central to ESG reporting because it identifies the most significant environmental, social, and governance issues that affect stakeholder decision-making and value creation for the company. Unlike adopting a one-size-fits-all strategy in ESG reporting, materiality enables organizations to focus on those issues relevant to their operational context, industry exposure, and stakeholder interests (Jiang et al. 2025). In the case of VSMEs, where capacities are constrained, materiality is not only a compliance necessity but also a strategic necessity for the optimization of ESG initiatives without overcapacity allocation. Artificial Intelligence (AI) offers a transformative approach to materiality mapping through its ability to handle huge-scale, multi-dimensional sources of data at an affordable and adaptable cost. Artificial intelligence-based materiality analysis software is now capable of better comparing a firm's ESG issues to industry-specific expectations, regulatory requirements, and investor sentiments. By incorporating structured information such as financial performance, carbon exposure, and employee composition, and unstructured information such as annual reports, stakeholder views, and media, AI has been able to detect materiality of issues more accurately and contextually.

One of the significant developments in this direction is the integration of stakeholder sentiment analysis. Natural Language Processing (NLP) technology is used in order to analyze text data from the likes of news articles, press releases, investor presentations, and social media. The models track issues rising around ESG themes as well as stakeholder expectations, allowing for a dynamic reassessment of materiality. For instance, if sentiment information reveals growing concern for supply chain emissions in a certain industry, AI systems can update the weighting of materiality for Scope 3 emissions accordingly. In this manner, materiality analysis is constantly updated, forward-looking, and aligned with real events. Machine learning (ML) also helps achieve this by designating materiality or non-materiality to ESG issues with supervised and unsupervised learning models. These learn from previous ESG disclosures, regulatory filings, and corporate sustainability reports across industries (Autto et al. 2024). Having been trained, the algorithm can

then examine a company's operational and stakeholder information to sort out what part of ESG is financially or reputational material. Classification is automatically refreshed in feedback loops, refreshing with new data in order to improve predictability.

Besides, AI platforms enable adaptive materiality that captures regional, firm size, and industry differences in ESG materiality. Water use materiality, for example, might be greater in water-stressed regions, and information security might be more material for companies with a high level of digital services. By integrating geospatial and sectoral information, AI platforms can develop customized materiality maps that adjust to the business environment. Such fineness of detail makes ESG reporting not just compliant but also operationally relevant to the reality it reflects. With the marriage of machine learning and AI, materiality assessment transitions from a static, periodic activity to a real-time, smart system reflecting shifting stakeholder expectations and dynamic regulatory landscapes (Kljaić *et al.* 2024). This intelligent automation not only enhances reporting accuracy and value but also allows VSMEs to perform high-quality ESG disclosure at zero cost or without requiring extensive sustainability expertise.

Component	Description	Relevance to VSMEs
Materiality Mapping	Identification of ESG topics most relevant to stakeholders and company value.	Helps focus resources on high-priority issues; avoids overreporting.
AI-Based Benchmarking	Comparison of ESG issues against industry norms and sector-specific benchmarks.	Ensures alignment with sectoral ESG expectations without manual analysis.
Sentiment Analysis	NLP algorithms analyze stakeholder sentiment from media, social platforms, and public filings.	Captures real-time shifts in stakeholder concerns without needing in-house teams.
ML Classification of ESG Issues	Supervised/unsupervised learning models classify ESG issues as material or non-material.	Reduces complexity and ensures consistency in determining what to report.

Adaptive Materiality	Adjustment of materiality priorities based on geography, sector, company size, and market trends.	Provides VSMEs with a dynamic, tailored materiality matrix instead of static templates.
Automated Updates and Feedback Loops	Continuous learning from new data to improve materiality recommendations over time.	Ensures reporting evolves with regulatory and stakeholder changes.
Data Integration (Structured + Unstructured)	Combines internal metrics with external qualitative sources (e.g., CSR reports, audits).	Enables holistic assessment with minimal manual data processing.

Table 2.7.1: AI-Driven Materiality Assessment Features in ESG Reporting

2.8 Comparative Analysis of Existing ESG Reporting Platforms

Growing ESG transparency demands have seen the development of various digital platforms aimed at helping organizations aggregate, analyze, and report ESG data. The platforms vary significantly according to technology capabilities, target users, ESG framework support, and how much artificial intelligence is integrated into them (Leivaditis & Sepetis, 2025). A comparative analysis of top ESG reporting platforms highlights the principal gaps and opportunities, most notably in terms of catering to the needs of very small and medium-sized businesses (VSMEs). A first category of platforms consists of enterprise-scale ESG software such as Sphera, Enablon, and SAP Sustainability Control Tower. These tend to be better suited for large multinational corporations with substantial reporting duties and data infrastructure. These offer extensive enterprise infrastructure integration, analytics at high levels, configurable reporting modules that are configurable, and multi-standard support (e.g., GRI, SASB, CDP, TCFD). These are typically high-cost implementation and usage solutions with complex user interfaces and high-resource configuration requirements. These are thus not feasible or available for VSMEs, who are frequently sans special ESG teams or IT. A second category is lightweight SaaS products such as Planmark.io, Sustain. Life, and FigBytes.



Figure 2.8.1: Comparative Analysis of Existing ESG Reporting Platforms

(Source: Xu et al. 2025)

These typically offer templated reporting capabilities aligned to certain frameworks, cloud-based data entry, and graphical dashboards. Their value lies in being simple to utilize, relatively lower prices, and their ability to create plain ESG reports relatively rapidly. However, most such tools are still limited by stiff data entry models and do not automate any of the significant activities, such as ESG metric extraction, materiality identification, or narrative reporting creation in full. A few platforms also come with semi-automatic calculators for emissions data, but these are less context-sensitive and adaptable than AI-powered systems. More recent platforms are beginning to incorporate artificial intelligence, particularly in natural language processing, predictive analytics, and real-time data monitoring. For example, some tools now use NLP models to review sustainability reports or auto-generate ESG report sections. Others offer AI-based materiality assessments that filter public opinion, peer benchmarking, and regulatory drivers to suggest relevant ESG topics (Ullah *et al.* 2025). However, these AI applications are still in early stages of application and are usually confined to specific segments of ESG, such as environmental footprint or carbon reporting.

Also, their algorithms are typically not open as proprietary and are non-transparent, and hence it is difficult to ascertain their accuracy or bias, a very important requirement in compliance and ethical reporting. There is one more serious shortcoming of the majority of platforms, that do not properly support the special needs of VSMEs. The VSME report model, nonetheless, as introduced by the European Sustainability Reporting Standards (ESRS), mandates a group of reduced

indicators and disclosure rules exclusively for small businesses. Consequently, available platforms do not have modules exclusively for VSME templates or instructions. This deficiency adds more loads to small firms looking to comply but cannot modify the general ESG tools to their activities and capacity. The comparative study indicates that although technology for ESG reporting has evolved significantly, there are no platforms with low complexity, high automation, real-time intelligence, and VSME compatibility (Zhu et al. 2024). A software that combines artificial intelligence for extraction, interpretation, and tailor-made report generation—despite the optimized ESRS VSME design—is the need in the current ESG reporting system.

Platform	**Target User**	**Frameworks Supported**	**AI Integration**	**Automation Level**	**VSME Compatibility**	**Notable Limitations**
SAP Sustainability Control Tower	Large Enterprises	GRI, SASB, TCFD, CSRD	Limited (mainly dashboards and analytics)	High	Not suitable	High cost, complex setup, steep learning curve
Enablon	Multinational Corporations	GRI, CDP, TCFD	Predictive analytics and scenario models	High	Not suitable	Requires integration expertise, expensive
Planmark.io	Small and Medium Enterprises	ESRS VSME (partial)	Minimal (rule-based assistance)	Moderate	Intended for VSMEs	Limited AI functionality, basic automation

Sustain. Life	SMEs and Startups	GRI, B Corp, Custom Metrics	Basic automation, no adaptive learning	Moderate	Suitable but generic	Limited ESG domain intelligence, manual inputs
Datamaran	Mid to Large Enterprises	GRI, SASB, CSRD	NLP for risk identification	Moderate	Not VSME-specific	Focused on strategic risk, less on full reports
Microsoft Cloud for Sustainability	Enterprises with Azure Stack	GRI, CDP, Custom Configurations	Dashboarding, data mapping	Moderate to High	Overbuilt for VSMEs	Requires Microsoft ecosystem, limited flexibility
FigBytes	Mid-sized Organizations	GRI, TCFD, UN SDGs	Limited AI, prebuilt templates	Moderate	Partially suitable	Lacks AI-driven insights, limited customization

Table 2.8.1: Comparative Analysis of Selected ESG Reporting Platforms

2.9 Literature Gap

Despite the growing research literature on ESG disclosure and artificial intelligence applications in sustainability analytics, the literature exhibits a huge knowledge void concerning the development of AI-aided ESG reporting tools tailored for Very Small and Medium Enterprises (VSMEs). Most existing studies, as well as business tools, focus on large enterprises with sound reporting infrastructure, thus missing smaller companies. Such firms typically face specific

challenges, e.g., limited financial and technological capacity, lack of internal ESG expertise, and difficulties with dealing with advanced reporting regimes such as the CARD or EARS. Furthermore, while the use of AI software such as Natural Language Processing and machine learning for handling ESG data and reporting has been addressed, there is negligible research into the usage of these technologies in end-to-end systems that are accessible to non-technical users. Few educational or industry research studies have investigated how AI can facilitate materiality analysis, automate metric extraction, and generate uniform ESG disclosures in the distinctive regulatory context of VSMEs. This thesis addresses this critical gap by proposing an end-to-end, AI-driven ESG reporting solution with integrated VSME frameworks. It is academically valuable as well as practically useful by emphasizing automation, accessibility, and compliance orientation in small business ESG reporting.

Chapter 3: Research Methodology

3.1 Research Philosophy

The philosophical underpinning of this study falls under the pragmatic method of inquiry, as it concentrates on responses that enhance the reporting of ESG through the incorporation of AI. Pragmatism is focused on practice and practicality, and this approach is suitable for studying the possibilities of including AI technologies in ESG reporting to confront the current problems of this reporting in practice. Such a philosophy has the opportunity to be flexible in the application of qualitative and quantitative data when measuring the efficacy of AI tools. This way of thinking about research findings and creating usable knowledge guarantees that this kind of study offers practical recommendations that are used by organizations that want to improve the accuracy, efficiency and real-time integration of their ESG reporting activities.

For ESG disclosure, embracing AI technologies has the potential to revolutionize, particularly for Very Small and Medium Enterprises (VSMEs) that cannot invest in comprehensive disclosure. A realist approach is supportive of investigating mechanisms through which AI could ensure accuracy, reduce labor, and enable immediate processing of data in ESG disclosures. This philosophical perspective allows for the conjoining of qualitative methods—i.e., expert interviews and case studies—with empirical investigation of AI platforms like Planmark.io. The research is therefore well-equipped to develop insights that are not only theoretically sound but also implementable (Chung & Latifi, 2024). The pragmatic philosophy further encourages responsiveness to emergent technology and stakeholder concerns. ESG reporting responsibilities become more and more influenced by regulatory evolution, investor activism, and public attention. Through its focus on outcomes with direct applicability to reporting practices, pragmatism ensures that the research contributes to developing practical methodologies and tools. It also supports iterative learning to enable the researcher to modify the framework in light of literature as well as practice-based evidence.

Pragmatism is not restricted to a single epistemological tradition and therefore supports an adjustable approach to constructing knowledge. This adaptability is essential in research areas where technology innovation intersects with compliance and governance, as in AI-powered ESG reporting. This approach allows the research to broach the technical, ethical, and operational dimensions of AI adoption within the context of VSME. Through the focus on actual usefulness and applicability in the real world, the pragmatic philosophy is aligned with the objective of the

study in designing a practical, AI-driven ESG reporting tool that can be customized for the specific needs of various types of organizations (Su et al. 2025). Pragmatism is thus the underlying worldview upon which the research problem is investigated, research methods are selected, and outcomes are interpreted to establish correspondence between research objectives and practical utility.

3.2 Research Approach

This thesis research follows an empirical research methodology, and major emphasis is given to the qualitative research approach. The paper presents the concept of the use of AI in "ESG (Environmental, Social, and Governance) reporting", examining the current frameworks such as VSME and how these are improved with the help of automation using AI. An extensive literature review is carried out in order to assess the modern state of "ESG reporting", the issues of data collection, and the implementation of automated processes (Anuar et al. 2025). The study engages case studies of organizations, which utilize ESG reporting tools, such as the Planmark.io platform, to determine the effectiveness of AI-powered solutions in terms of increased accuracy, efficiency, and dynamism of data integration. The interviews with industry experts are also in the research process to obtain information about technical, data-related, and ethical issues of turning to AI-assisted ESG reporting. The given approach allows having a comprehensive picture of the possibility of AI streamlining the ESG reporting procedures and improving decision-making.

Theoretical Grounding through Literature Review

The foundation of the research is a thorough review of published academic literature, industry white papers, and regulatory filings regarding ESG reporting standards, AI integration, and how VSME compliance mechanisms can be established. This stage enables the recognition of key reporting issues such as fragmented data sources, inconsistent disclosure practices, and insufficient automated processes. Having defined this theoretical benchmark, the research is then able to critically analyze how AI-based solutions—natural language processing, data mining, and adaptive analytics—yield significant improvements. The literature also assists in benchmarking what is currently available and recognizing technical gaps that still exist and are not being filled by current ESG platforms.

Applied Analysis through Case Studies

Case studies are undertaken through empirical investigation of organizations that make use of AI-based ESG reporting systems, with particular emphasis on Planmark.io. The case studies

investigate the working mechanism of the tool, its applicability in the automation of ESG data, and compliance with the VSME principle. Actual data is gathered related to the usage of the system, data integration, report efficacy, and promptness (Rudd et al. 2023). This research aspect offers insight into the functional impact of AI tools and scales through organizational size and industry sectors. The case studies also indicate issues with implementation, including interoperability, algorithmic transparency, and user adoption barriers.

Expert Insights through Qualitative Interviews

Apart from testing hypotheses and placing the case study findings, the study entails semi-structured interviews with ESG professionals, AI software engineers, and sustainability consultants. The interviews seek to gather nuanced perspectives on the ethical, technical, and regulatory dimensions of AI integration in ESG reporting. The interview data undergoes thematic analysis to explore recurring themes, industry differences, and forward-looking recommendations. This qualitative approach enriches the empirical analysis of the research and enables it to provide policy-makers and technology creators with relevant insights to develop AI systems that meet evolving ESG disclosure norms.

3.3 Research Design

The study design of the question under analysis is descriptive-exploratory, where the researcher seeks an explanation of the integration of the “AI” into the “ESG reporting” and, in particular, on the VSME reporting model and its application on the Planmark.io site. The design uses qualitative and empirical methods to investigate the ways of improving the ESG reporting procedures with AI; in particular, in terms of accuracy, efficiency, and real-time integration of data. The research is characterized by three key stages that include literature review, case studies, and expert interviews.

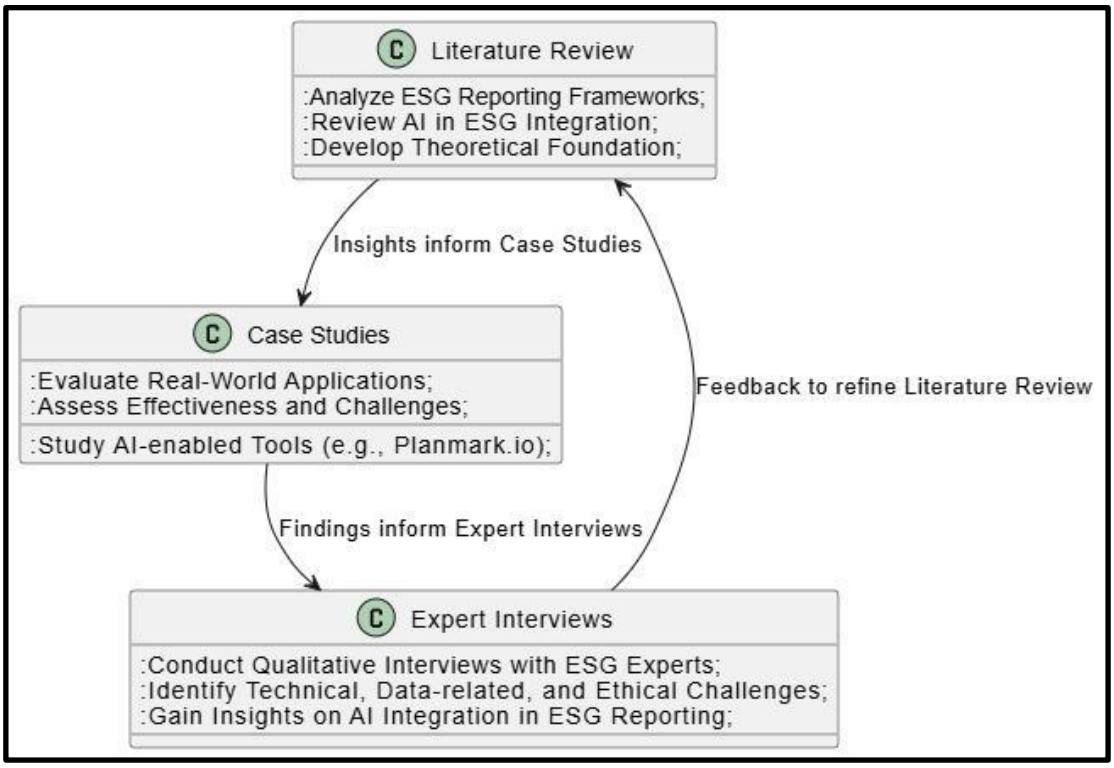


Figure 3.3.1: AI-Assisted ESG Reporting Design Flowchart

Literature Review: The initial step includes a close analysis of the studies that have been conducted on ESG reporting models, in this case, VSME. The review examines the existing situation relating to ESG reporting, issues experienced by organizations, and the possible contribution of AI in solving them (Katterbauer & Moschetta, 2022). The theoretical background of AI-enabled ESG reporting is designed with the help of key academic materials and industry reports.

Case Studies: The second step involves the case studies of the companies and organizations applying such ESG reporting tools as Planmark.io. These are concentrated on learning the functionalities of the platform, especially AI-based capabilities, including automation in data collection, the incorporation of real-time data, and the creation of actionable knowledge. Case studies are done to determine the effectiveness and difficulties of these tools when applied in real life.

Expert Interviews: In the last stage, the interviews with ESG experts are planned to collect testimonies about practical difficulties in implementing AI in ESG reporting. These professionals give feedback on such tools as Planmark.io and share their experiences concerning technical, data-related, and ethical issues of generating the switch to AI-assisted reporting.

Phase	Description	Tools/Methods
Literature Review	Analyze existing ESG reporting frameworks and AI integration	Academic papers, industry reports, theoretical foundation
Case Studies	Evaluate real-world applications of AI tools (e.g., Planmark.io) in ESG reporting	Case study analysis, Planmark.io platform
Expert Interviews	Collect insights from ESG professionals on AI integration in reporting	Semi-structured interviews, qualitative analysis

Table 3.3.1: Summary of Research Design

(Source: Yang, 2025)

The research design is iterative, such that the findings of the literature review and case studies present in the expert interviews, making them create a complete picture of the role of "AI in ESG reporting".

Case Study Phase

The second phase comprises several case studies of ESG reporting platforms, namely Planmark.io. The case studies offer tangible evidence for the application of AI tools to practices of sustainability disclosure. The case selection is based on their implementation of AI capabilities such as real-time data integration, automated mapping of compliance, and ESG visualization based on dashboards. The purpose of this phase is to determine the functionality and challenges in developing AI features and their compatibility with reporting requirements for VSME-specific reports. The results of this phase guide the pragmatics of tool design and decide areas where functionality can be improved.

Expert Interview Phase

The third stage consists of semi-structured interviews with experts in the ESG, AI development, and regulation areas. The interviews are aimed at confirming information gathered in the literature and case study stages and gathering firsthand opinions on working, ethical, and technical challenges (Gwon et al. 2024). Experts are also requested to provide opinions on system architecture, AI explainability, and stakeholder trust. Their research is thematically analyzed to determine best practices and cautionary issues relevant to the use of AI-based ESG report tools in the VSME setting.

3.4 Data Analysis and Collection Methods

Primary and secondary data are used as a combination in collecting and analyzing data in this study on AI-supported ESG reporting.

Secondary Data Collection: The researchers start by initiating a thorough literature review, including scholarly articles, industry publications, and other ESG systems (including VSME). Such a secondary source of information allows building a theoretical background of AI in the context of "ESG reporting", including defining existing issues and constraints, as well as tools, such as Planmark.io (Liao et al. 2023). The secondary data is in the form of qualitative analysis to know about the current methodology, its success and any loopholes in the present system.

Primary Data Collection

Case Studies: Real-life case studies of companies that use ESG reporting tools such as Planmark.io are reviewed. Evidence about the ESG reporting process, which concerns the integration of AI tools, challenges to be addressed, and increasing accuracy and efficiency, is gathered.

Expert Interviews: Semi-structured qualitative interviews are carried out with experts in the field of ESG, developers of AI programs and industry (Ghosh, 2025). The interviews address their views on the AI integration to ESC reporting and give answers about technical, ethical, as well as data-related issues. The thematic analysis of the data is performed to define the similarities in the data, trends, and suggestions made of AI-enabled ESG reporting. Compounding both the secondary and primary data, such a study offers an effective insight into the implications of AI on the ESG reporting systems.

This study takes a dual-source data methodology by mixing both primary and secondary data for examining AI adoption in ESG reporting. Secondary data is collected via a systematic search of peer-reviewed scholarly journals, regulatory reports, and industry magazines. This background research helps in developing a theoretical framework and in exploring current issues in ESG compliance, particularly among VSMEs. It also helps in finding existing AI-based ESG tools like Planmark.io.

Primary data is gathered through case studies and interview feedback from subject matter specialists. Case studies revolve around entities that employ AI-enabled ESG platforms, with Planmark.io serving as the point of reference. The case studies examine specific functionalities of tools such as automated data gathering, live metric importing, and accessibility for interfaces. Semi-structured interviews are conducted with ESG specialists, AI technology innovators, and

compliance specialists in order to gain rich information about the challenges of implementation, ethics, and technological viability of AI integration (Zhao & Gómez Fariñas, 2023). Thematic analysis is applied for data analysis, which equates to coding qualitative responses for patterns, recurring themes, and success factors. This method enables the derivation of advanced insights into how AI promotes the precision, effectiveness, and scalability of ESG reporting processes, particularly for resource-poor VSMEs.

3.5 Ethical Consideration

This study follows the research ethics to have integrity and transparency. The participants of the expert interviews sign an informed consent; thus, these know what the study is all about, and these are also entitled to confidentiality. The identity of organizations used in the collection of data is anonymized in the case studies consideration. Data safety is also guaranteed because all the data collected is stored on a secure platform, either during an interview or during a case study. Where required, ethical approval is sought, especially when it is supposed to access data that is sensitive to the organization. In addition, the study is not involved in any conflict of interest, and it maintains fairness during data analysis and its interpretation.

The work of research is conducted in strict observance of professional and academic ethics. All research activities involving human subjects, such as expert interviews, are anchored on ethics of justice, autonomy, and beneficence. Informed consent is openly obtained from all interviewees to ensure that participants are aware of the intent of the research, the reason for which their contribution would be utilized, and the voluntary nature of the study. Participants are also informed that can withdraw at any moment without penalty. Confidentiality is guaranteed by anonymizing organizational and personal identifiers, and data are stored securely on encrypted and password-protected digital systems.

Where case studies rely on specific organizations, sensitive or proprietary data are treated sensitively, and permission is sought where it is required. As a means of obviating prejudice and offering transparency, the research does not engage in any activity that attracts a conflict of interest (Pranesti & Romadhon, 2025). The data collection, analysis, and interpretation are carried out in a premeditated and objective manner. University policy also seeks ethical clearance from the concerned institutional authorities, especially where there are third-party external stakeholders in primary data collection. The tracking of these measures have been prevent violation of the rights

of the participants in the research and the credibility, reliability, and academic integrity of the research process.

3.6 Summary

The methodology includes a “literature review” with “case studies” and “expert interviews” to investigate the use of “AI in ESG reporting”. The theoretical indicators are offered with the help of secondary data, whereas primary data from real-life applications and professional opinions evaluate the effect that AI has on accuracy, efficiency, and integration of data, both in ESG reporting, in real-time.

In this chapter, a conceptual and empirical research procedure is highlighted, which attempts to find out how artificial intelligence is used in ESG reporting about the VSME system. The three key elements incorporated in the method are the literature review with the intention of determining theoretical background, case studies with the objective of examining practical application of plans like planmark.io, and professional interviews with the aim of collecting qualitative information on technical, ethical, and functional aspects. The use of primary and secondary data allows examining the issue of AI technologies to enhance the accuracy, effectiveness, and promptness of ESG reporting systems in detail. With the help of the descriptive-exploratory design, it is possible to describe the modern constraints and opportunities in the sphere.

Chapter 4: Research Findings

4.1 Introduction

In this project, an AI-driven ESG (Environmental, Social, and Governance) reporting system shall be developed and used in order to increase transparency and compliance with sustainability by small and medium-sized enterprises. Due to the emergence of regulatory frameworks such as the EFRAG VSME standard, companies have growing pressure to present factual, timely, and uniform ESG disclosure. Machine learning and natural language processing are used to complete the project by providing an online option to automate the data collection, risk detection, and finalising the report. This not only saves manpower and errors in reporting but also enables organizations to monitor performance, ESG gap, and ESG and strategic alignment with the environment and social responsibility.

This chapter introduces the system's outputs under the three ESG pillars, highlighting important performance indicators like governance compliance, waste management, greenhouse gas emissions, biodiversity footprint, diversity of the workforce, training, and work health and safety. Graphical information and dashboards produced by the system are also considered to demonstrate its operational capabilities and reporting level of detail. These are organized into thematic sections highlighting important ESG results and followed by a comprehensive review of the system's operational performance, pitfalls, and strategic repercussions. The book concludes with an observation of how AI-enhanced ESG tools can assist in narrowing prevalent gaps in sustainability reporting by small companies, not only to regulatory adherence but also to improved internal decision-making and value creation over the long term. The succeeding sections critically analyze these results and interpret their relevance in the overall context of sustainable business processes for VSMEs.

4.2 Result Analysis

Governance

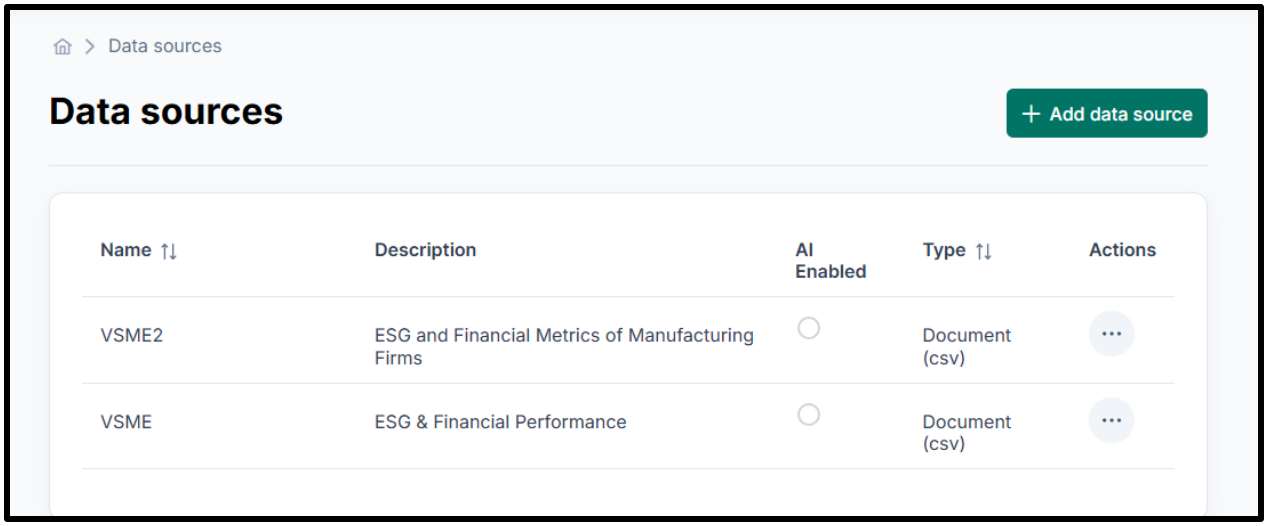


Figure 4.2.1: Data Sources for VSME Reporting

This figure provides the list of data sources of AI-based reporting of VSME reports and includes such structured datasets as CSV VSME2, VSME, with the names "ESG and Financial Metrics of Manufacturing" and "ESG and Financial Performance resources", which are used as input for automated sustainability analysis reporting and disclosure processes.

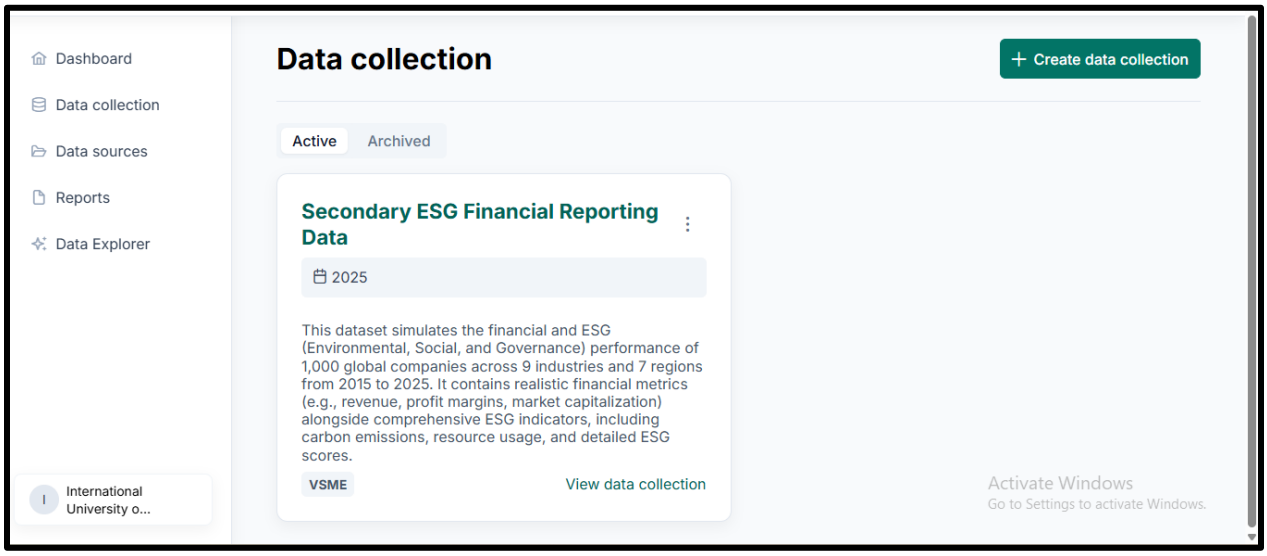


Figure 4.2.2: Data Collection Window

The Data Collection Window, as shown in this figure, is used by the users to access and manage ESG financial datasets to aid in the reporting using the AI feature (Zheng et al. 2024). It facilitates well-structured input of data, validation, and dynamic updating of information to provide efficient cross-industry, cross-regional, and cross-sustainability measures.



Figure 4.2.3: Data Collection View with Chapters

This figure represents the Data Collection View completed with ESG chapters, which indicates perfect compliance in such primary areas as Energy, Emissions, Biodiversity, and Pollution. This interface indicates how this AI system can monitor, control and approve ESG disclosures with ease.

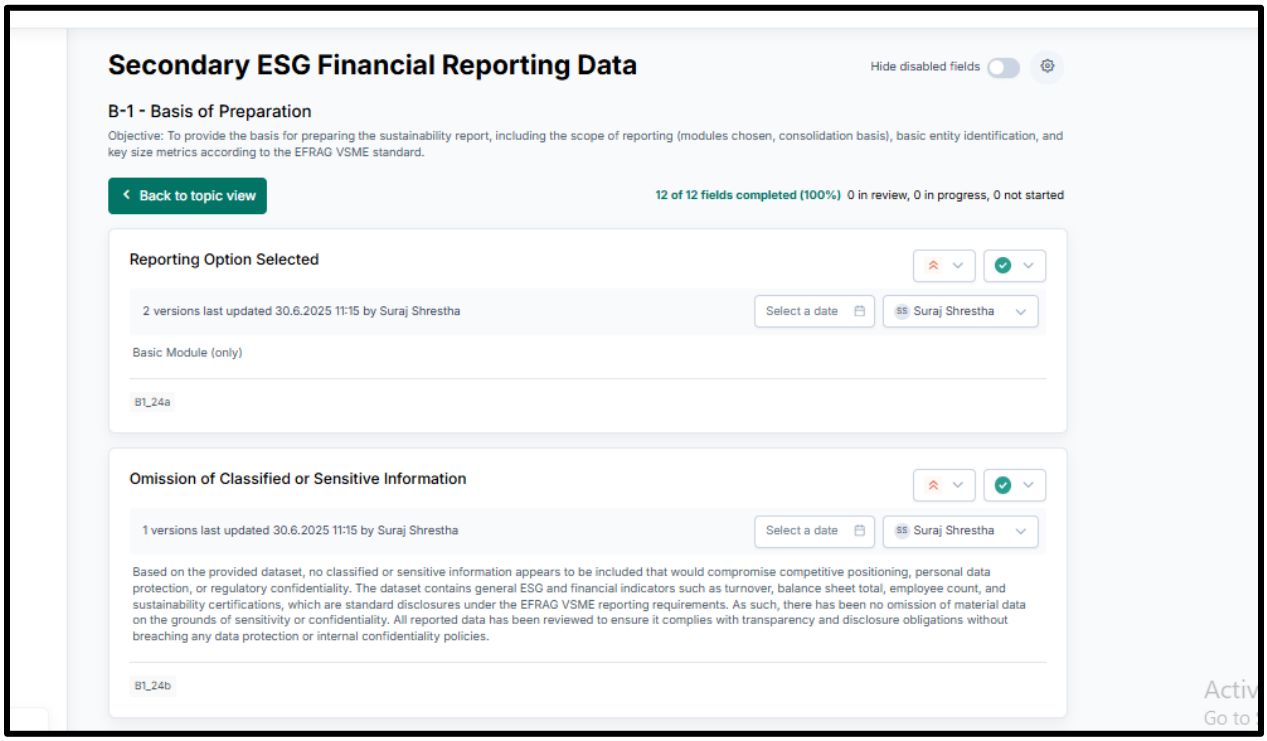


Figure 4.2.4: Basic Sensitive Information for The ESG Financial Report

This figure depicts the request of the ESG Financial Report with regards to the Basic Sensitive Information, which contains the details concerning the security of classified information, the ability to avoid reporting certain content and the precision of disclosures based on standards of EFRAG VSME and regulatory provisions.

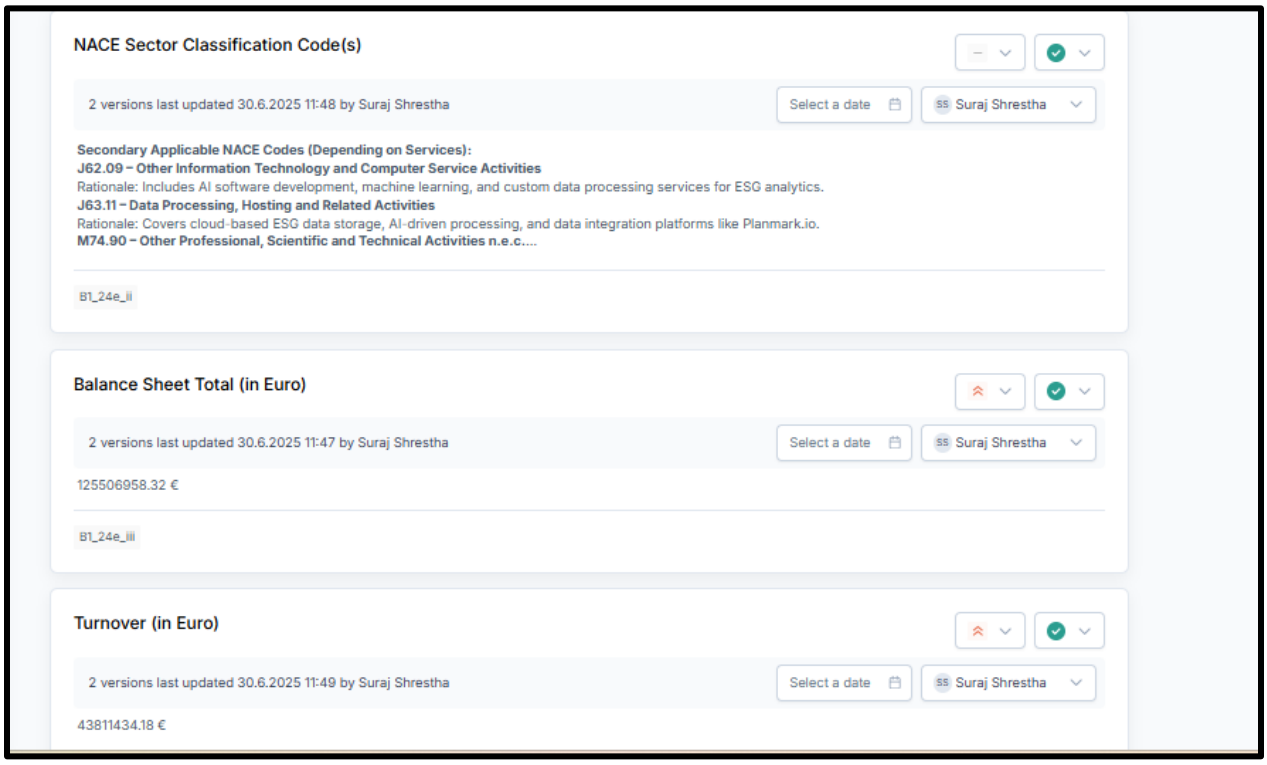


Figure 4.2.5: Balance Sheet and Turnover in Euro

Holds a balance sheet capital of 125506958.32 Euros and an annual turnover of 43811434.18 Euros. These financial key figures are needed to know the GHG intensity and identify the size of organizations to meet the criteria of the VSME category of AI-driven ESG assessment.



Figure 4.2.6: Circular Economy Principles for Waste Management and Resource Usage

This figure shows the circular practices implementation and indicates that 7.6 out of the 8.2 tons of non-hazardous waste have been diverted to be able to be reused or recycled, which constitutes 92.7 per cent of a circular recovery rate that has been optimized by the AI with a focus on finding efficient use of resources.

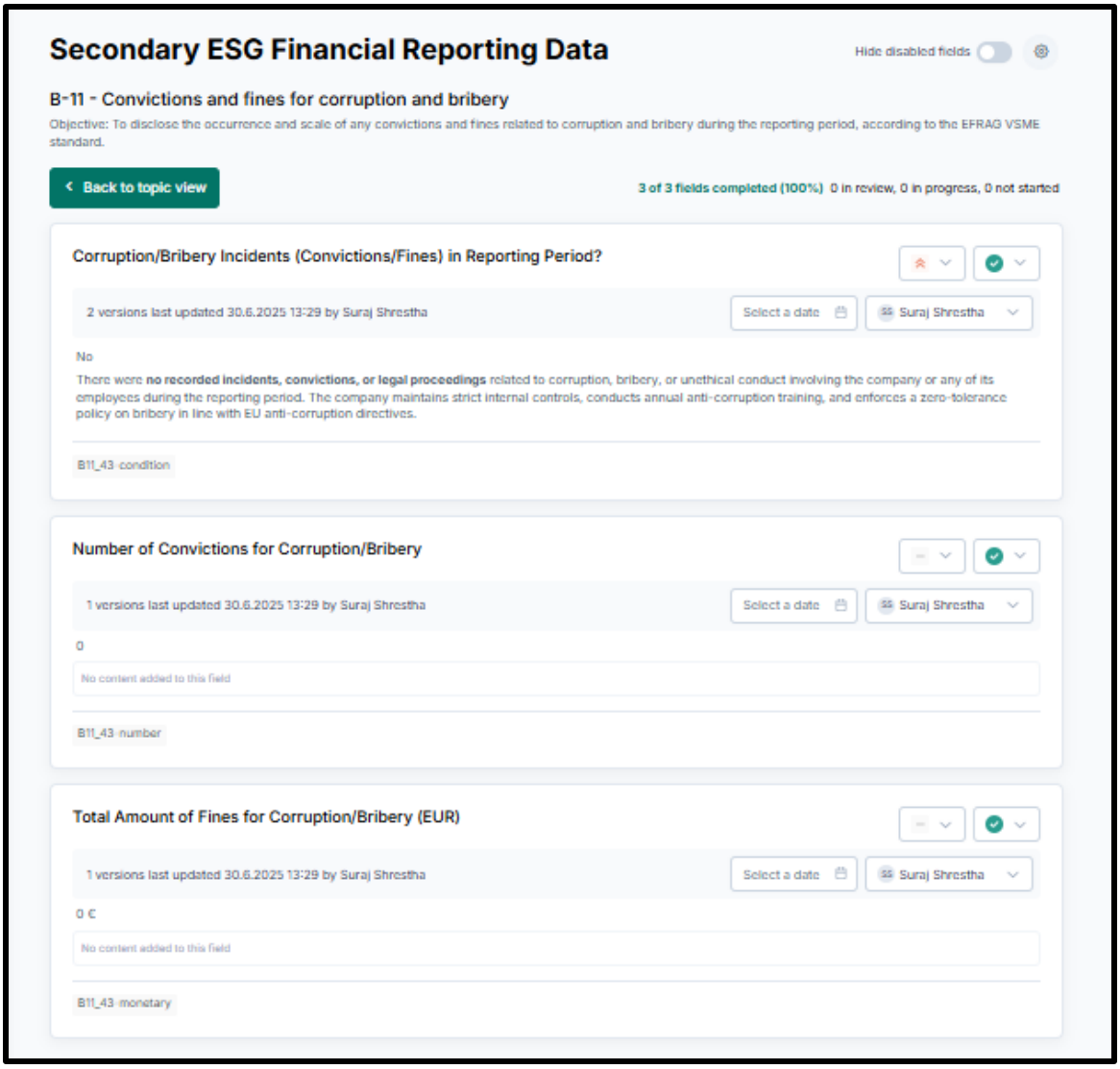


Figure 4.2.7: Corruption Analysis for ESG Reporting

This certifies zero punishments and EUR0 fines for corruption or bribery in the reporting year (Bock et al. 2025). Ethics guidelines and anti-bribery procedures are checked by AI tools that facilitate a clear ESG report and governance.

Environmental

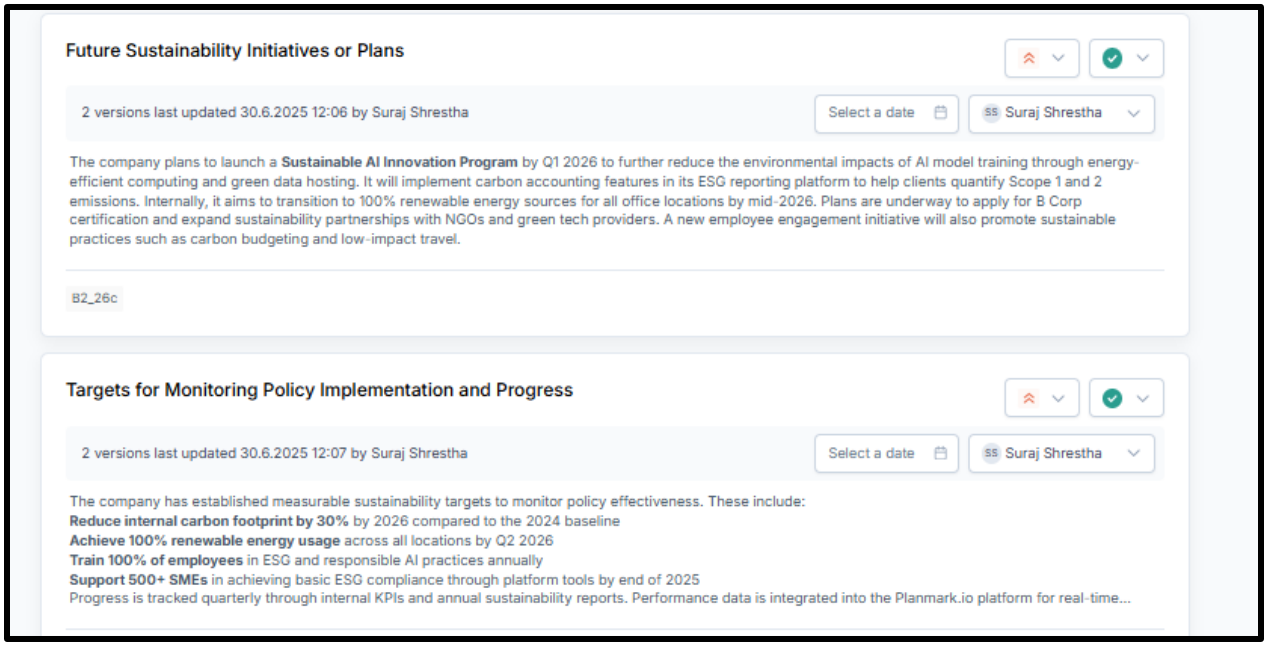


Figure 4.2.8: Future Sustainability Initiatives and Monitoring Targets

This figure mentions some of the AI-tracked goals, such as decreasing the amount of energy consumed annually by 10% and making sure that the recycling quota reaches at least 95%. Efforts being made are the growth of remote work, the use of green procurement, and ISO-certified propaganda of e-waste transactions.

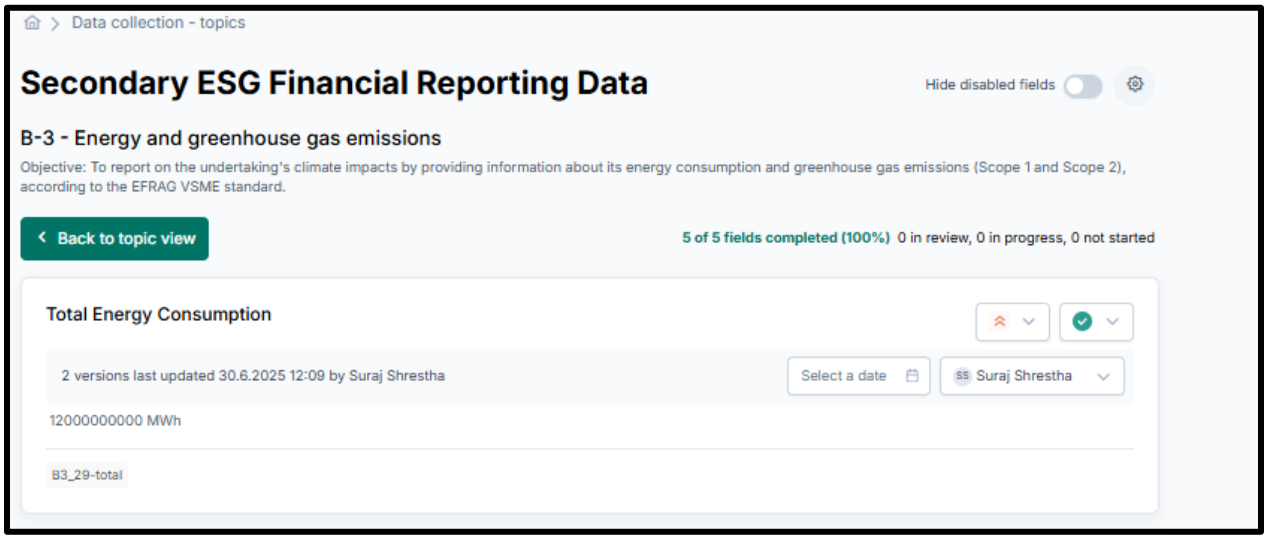


Figure 4.2.9: Total Energy Consumption

It reports 120,000,000,000 MWh of energy use in electricity as well as fuels. The AI analysis divides it into renewable and non-renewable categories in favour of energy efficiency and carbon tracing. The AI-powered ESG tool operates on this energy data through the segregation of

consumption into renewable and non-renewable consumption, thereby making it simpler to achieve increased carbon-intensity visibility about the utilization of green energy. This segregation assists in strategic energy management by enabling real-time monitoring of carbon, detection of high-consumption units, and energy reduction possibilities.

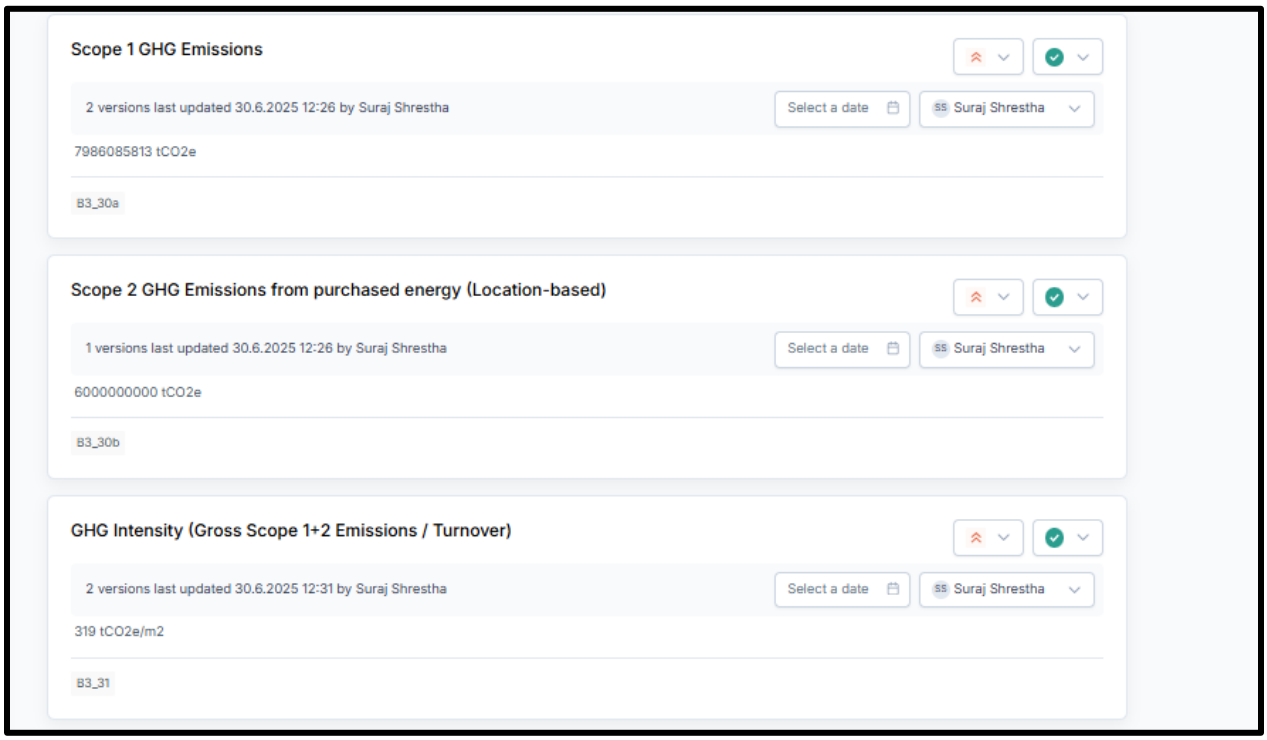


Figure 4.2.10: GHG Emissions and Intensities

This figure displays Scope 1 and Scope 2 emissions of 7.99 billion tCO2e and 6.00 billion tCO2e, respectively (Oleksandr and Haozhe, 2025). AI-based GHG intensity is 113.7 tCO2e/1M turnover. Indicators of climate impact per revenue are benchmarked. Scope 1 emissions are 7.99 billion tCO_2e, and Scope 2 is 6.00 billion tCO_2e. The AI system calculates a GHG intensity of 113.7 tCO_2e per €1 million turnover, an essential key performance indicator for climate footprint.

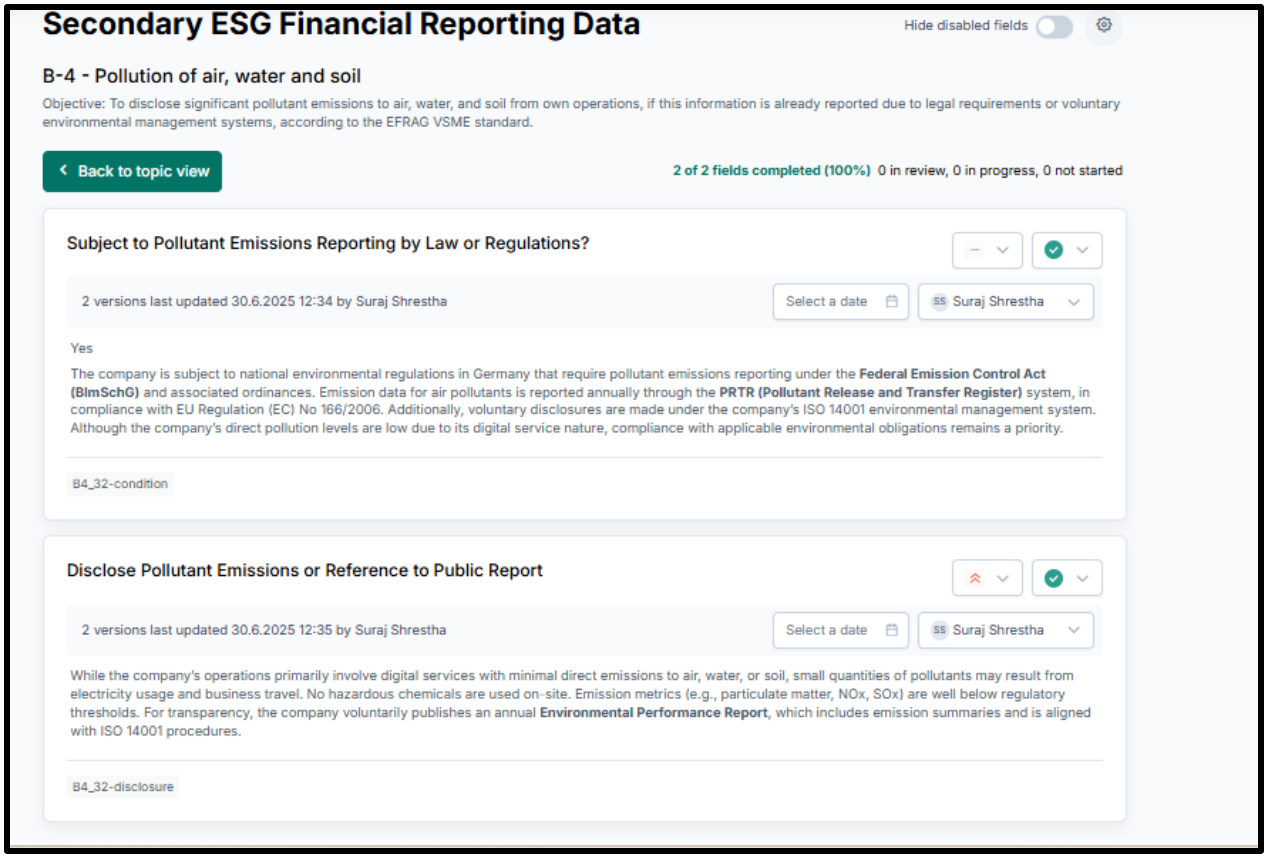


Figure 4.2.11: Details on The Pollution of Air, Water, and Soil

There are no reported emissions of pollution because of non-industrial activities. No legal reporting or voluntary disclosures associated with national environmental acts are identified in AI compliance scans. The AI software screens environmental compliance against environmental standards such as country-specific emissions regulations and EU directives and finds no breaches of the law or unreported pollution announcements. The platform verifies working and location data to identify any unannounced emissions threat and, in this case, found no pollution liabilities. This is further support for the low environmental risk nature of the firm.

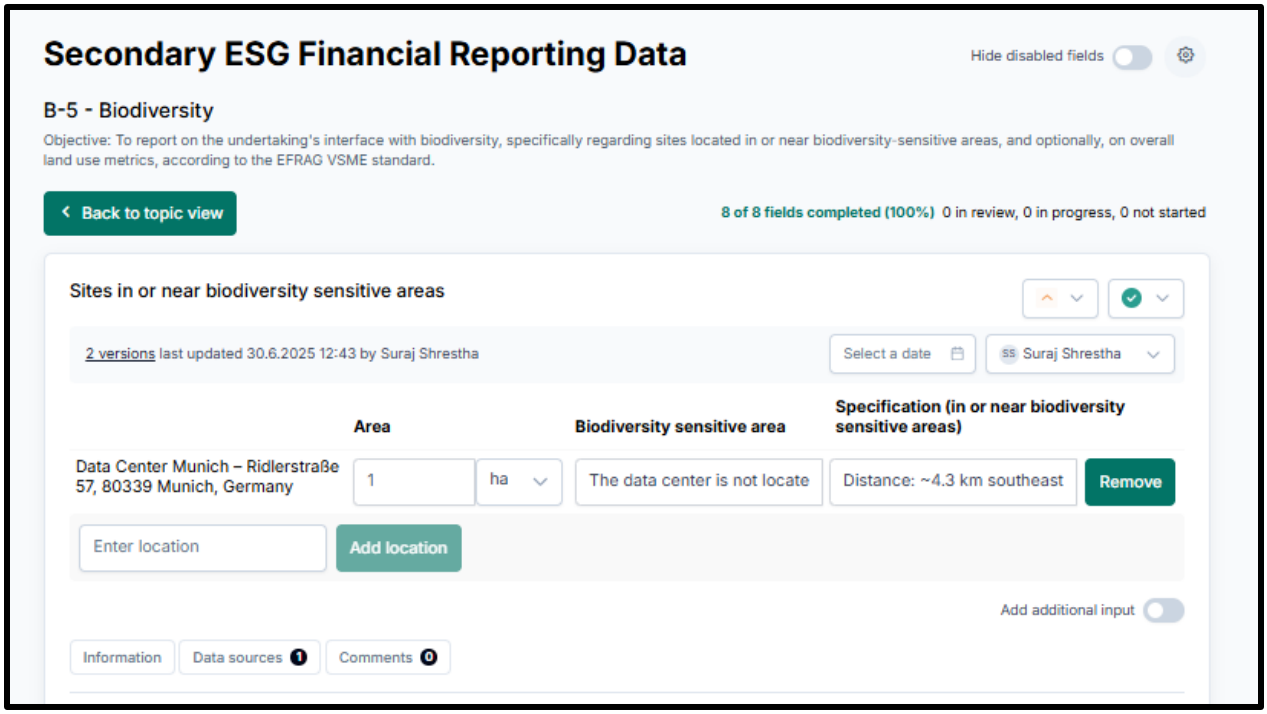


Figure 4.2.12: Specification of Biodiverse Sensitive Area

The AI-mapped operations sites show no direct sites within biodiversity-sensitive areas, albeit a single data centre that is 4.3 km away in the green corridor along the Isar River, in Munich, under observation, due to possible effects. The system has detected one data centre approximately 4.3 kilometres from the Isar River green corridor in Munich. Although not within a protected region, its relative location is in the running for inspection for the potential of ecological disruption.

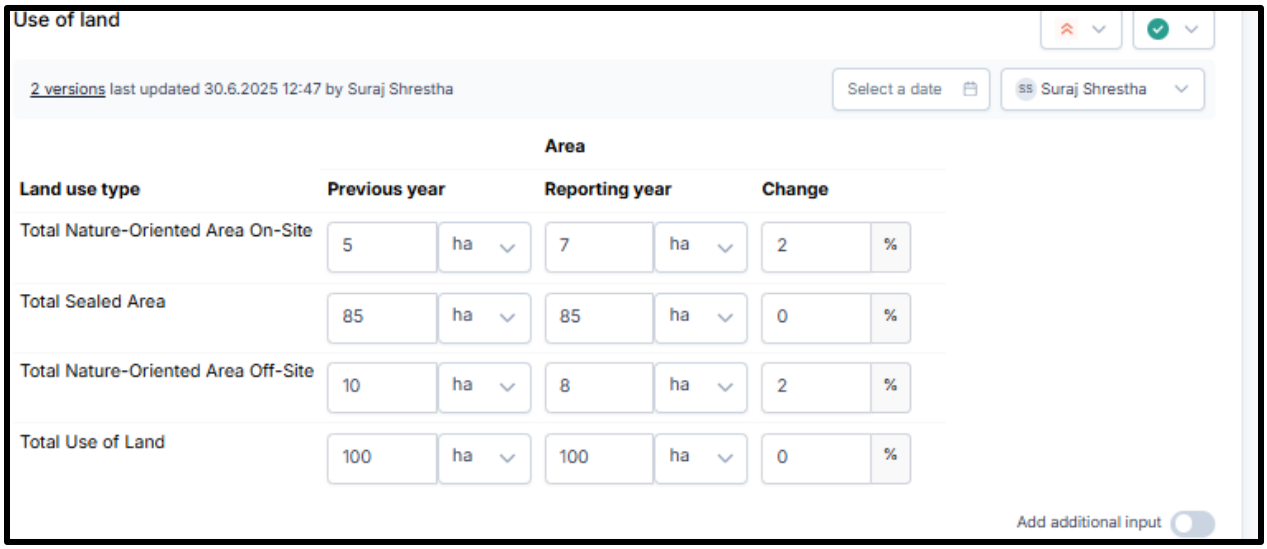


Figure 4.2.13: Total Use of Land to Maintain Biodiversity

This shows the cover of an area of 100 hectares of working land. AI assesses the situation in regards to land allocations and balances it with built-up (85%) and green (15%) environments according to urban biodiversity ideals (Celestin & Gidisu, 2025). These land use dynamics are assessed on urban biodiversity indicators, where AI models suggest optimization measures such as the introduction of indigenous plant species, reducing impermeable surfaces, and the extension of ecological corridors.

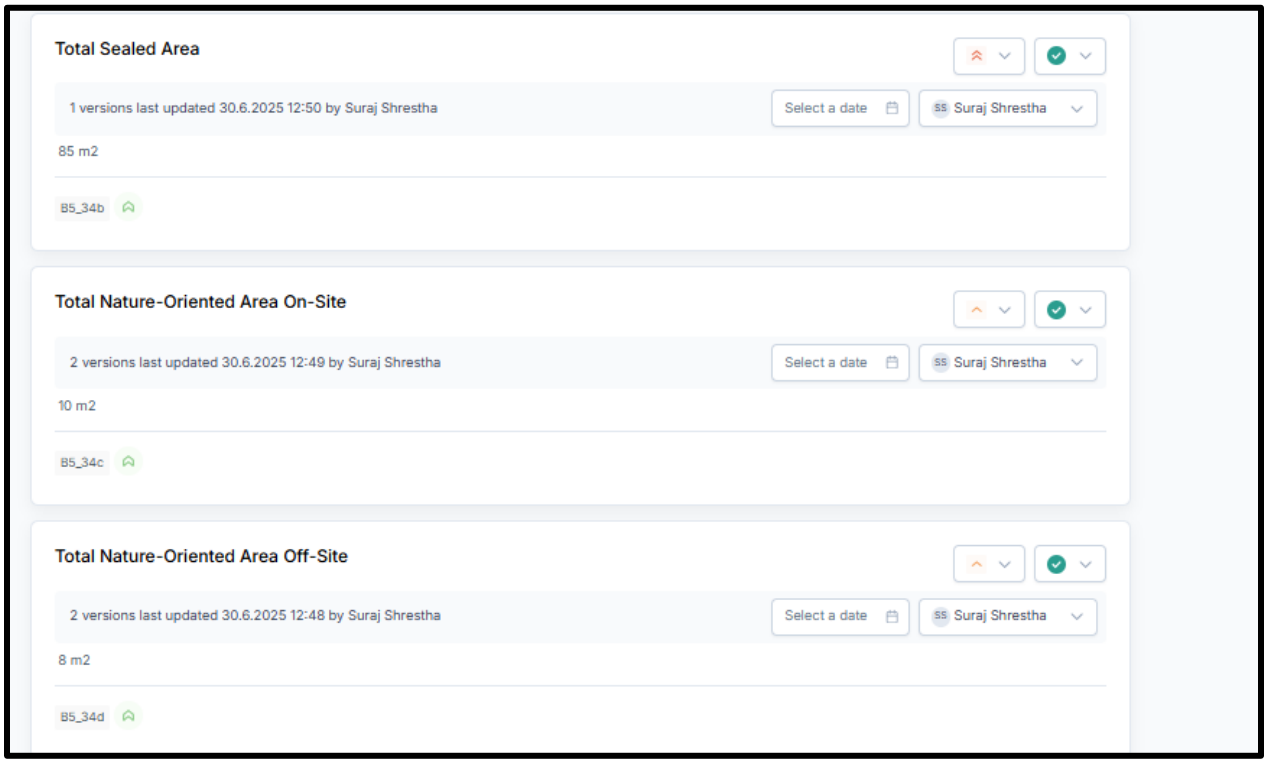


Figure 4.2.14: Sealed and Nature-Oriented Area

Land is 85 per cent (0.85 ha) sealed; 7 per cent (0.07 ha) nature-oriented on site and 8 per cent (0.08 ha) off site. AI monitors them to assess biodiversity scoring. The AI system aggregates this data to compute biodiversity scores and compare them to environmental thresholds. The platform also identifies when sealed land passes into levels that can have negative impacts on surrounding ecosystems, and recommends enhancements to green infrastructure, such as rooftop gardens, permeable pavements, and urban forest cover.

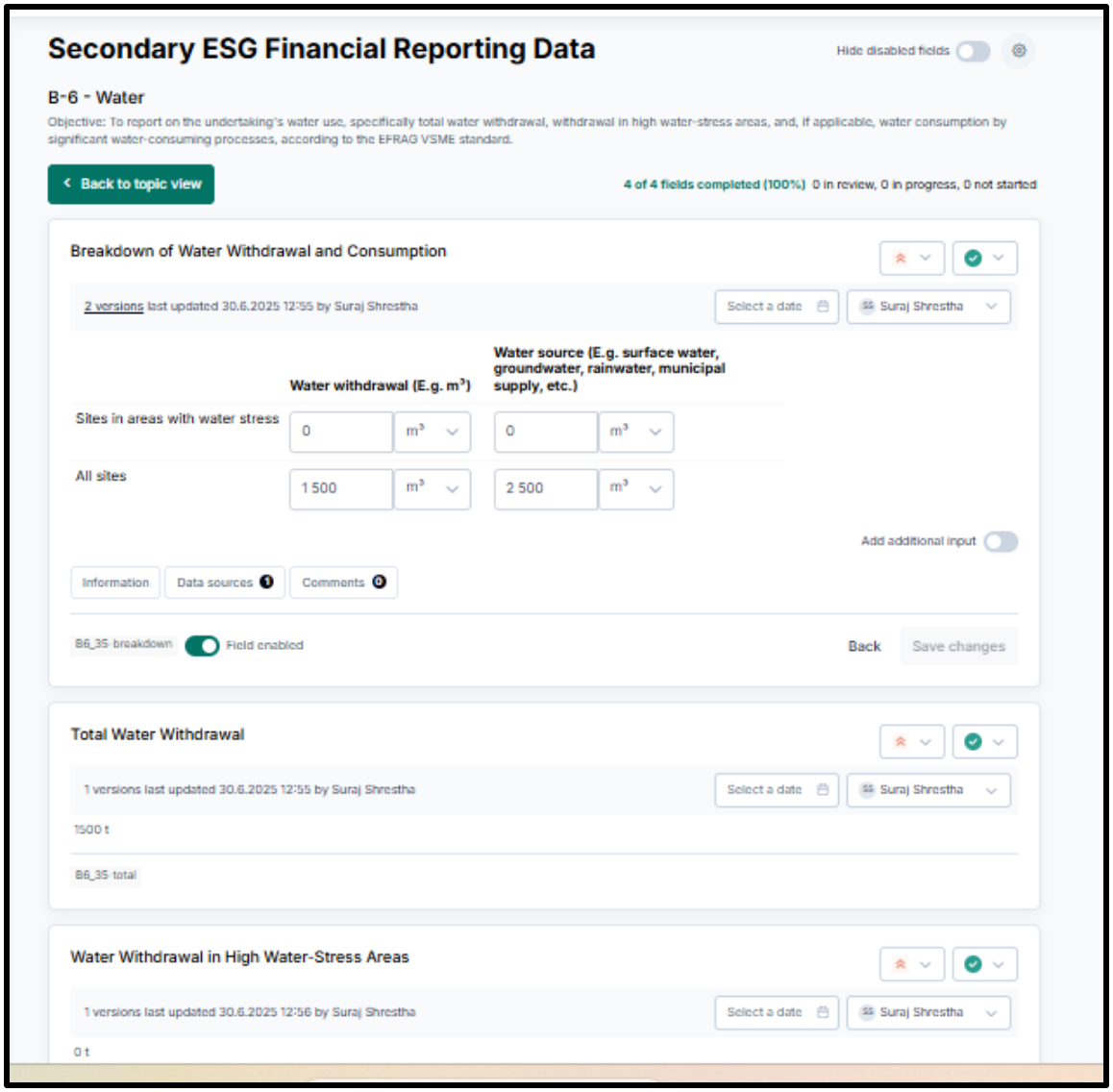


Figure 4.2.15: Water Withdrawal Specifications

The total water consumption amounts to 1,500 m 3 /a (1,500 tones/a) and all of it is municipal water. AI checks zero withdrawal out of high-stress water regions and zero water-intensive operations. Besides, the system notifies of any abnormal or excessive usage patterns, promoting healthier consumption behaviors. Absence of any water-consumptive procedures means a negligible hydrological footprint, the protection of which is critical towards environmental preservation in urban operations.

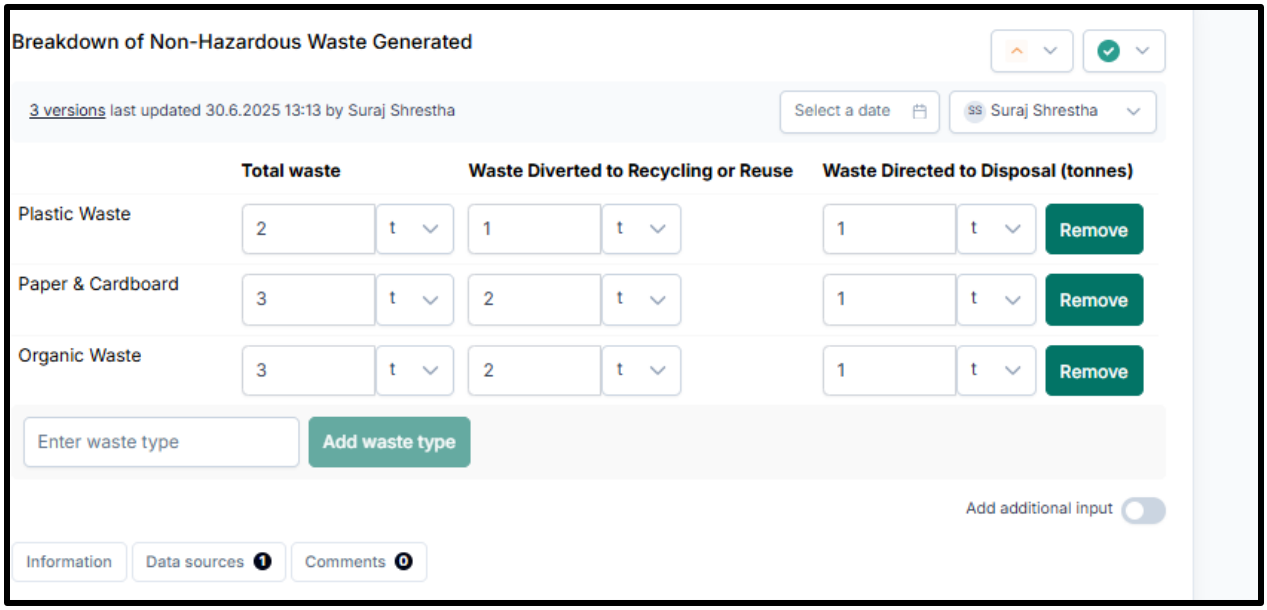


Figure 4.2.16: Non-Hazardous Waste Breakdown

Within the 8.2 tones, 3.5 tones comprises paper, 1.2 tones organic wastes and 2.5 tones plastic. AI categorizes and monitors the diversion effectiveness, 92 per cent recycled. Approximately 92% of this waste has been kept out of landfills effectively, demonstrating a high rate of circularity in line with European Union recovery objectives for waste. This is indicative of the success of digital monitoring in the promotion of circular economy conduct.

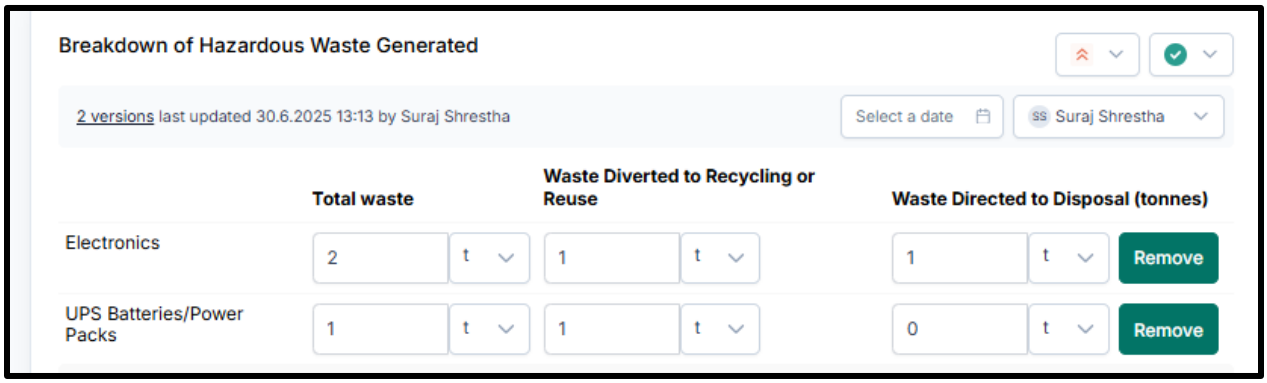


Figure 4.2.17: Hazardous Waste Breakdown

The quantity of total risky releases is 0.15 tons, of which 0.12 tons of electronic elements and 0.03 tons of UPS batteries. AI guarantees that it is certified and there is no landfill incidence. The AI technology ensures that all the dangerous substances are well-certified for safe handling and disposal in line with occupational and environmental health regulations (Kang et al. 2024). Remarkably, landfill toxic waste disposal has not been practiced anywhere, showing total adherence to optimal practice in toxic waste management.

Social

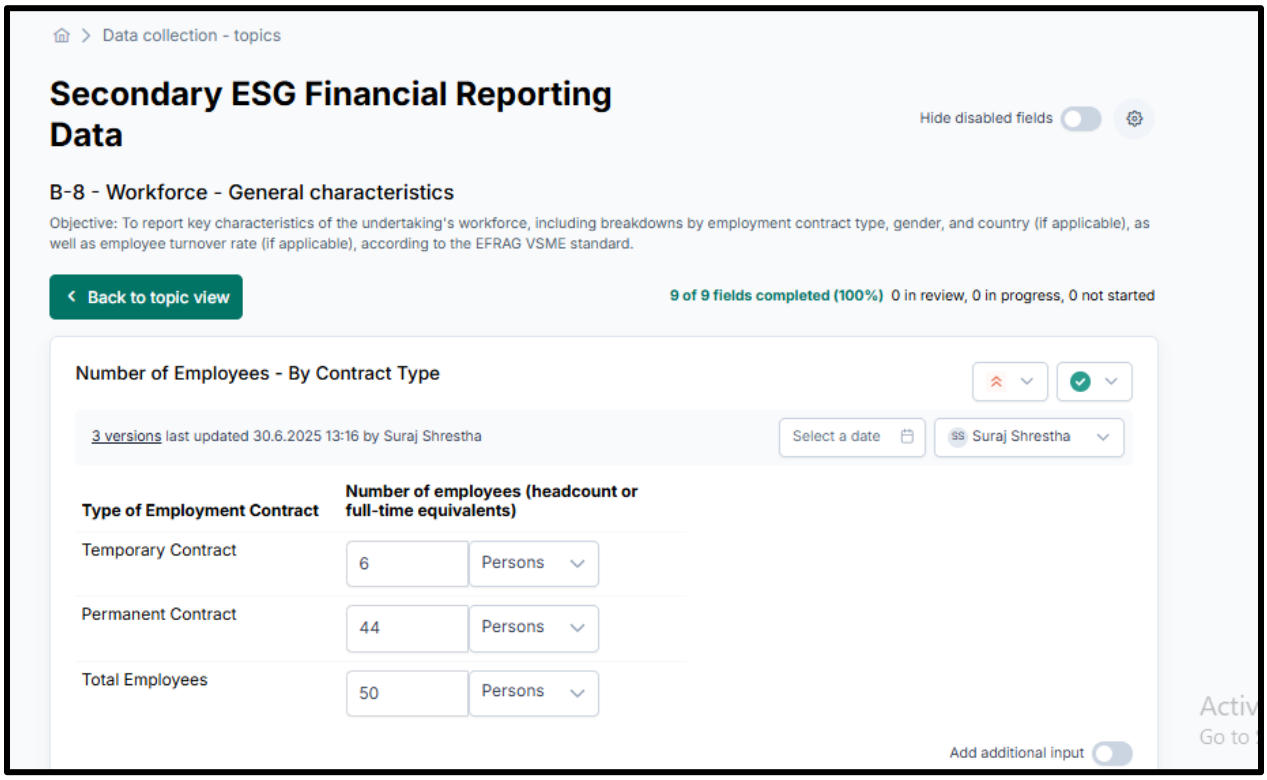


Figure 4.2.18: Contractual Workforce

This figure shows 6 workers under temporary employment and 44 under permanent employment. AI monitors retention and range of contracts to uphold equal labour policies. AI-based HR analytics continuously scan contract types to ensure equality, transparency, and conformity with labour law. The high permanent employee proportion (88%) reflects job security, and AI-driven insights influence workforce planning and retention plans.

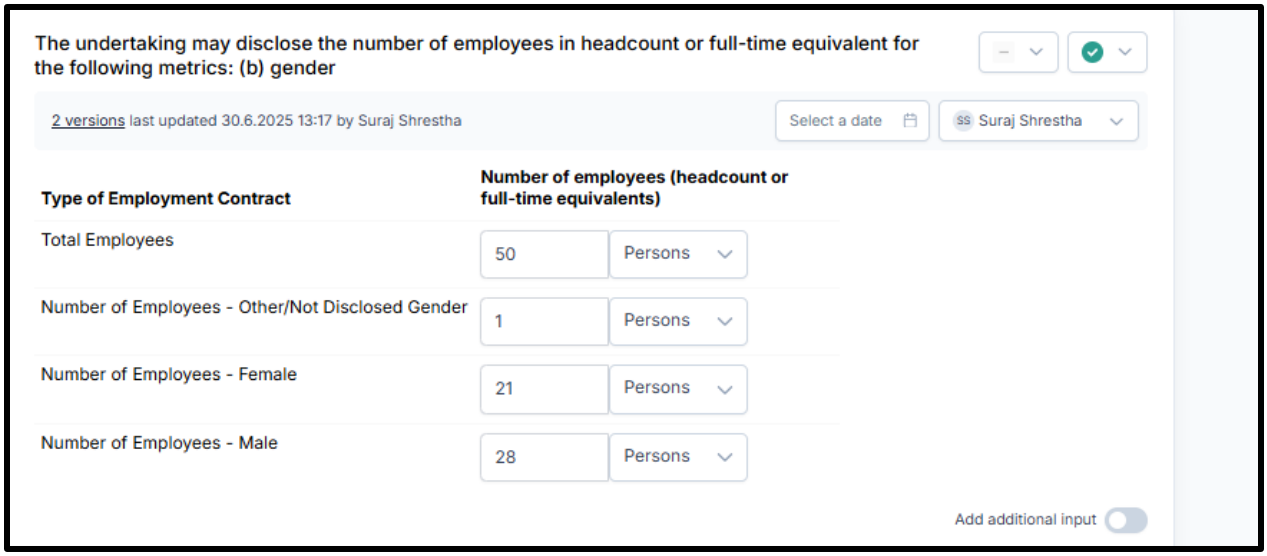


Figure 4.2.19: Gender-wise Workforce

The analysis shows 28 male employees, 21 female workers, and 1 unknown employee. AI measures the factors of diversity and ensures gender balance in hiring staff/ promotions. The AI tool keeps track of gender representation across levels of management and departments, enabling examination of the fairness of recruitment and promotion practices. This method improves the overall ESG mandate of diversity and inclusion workplace culture.

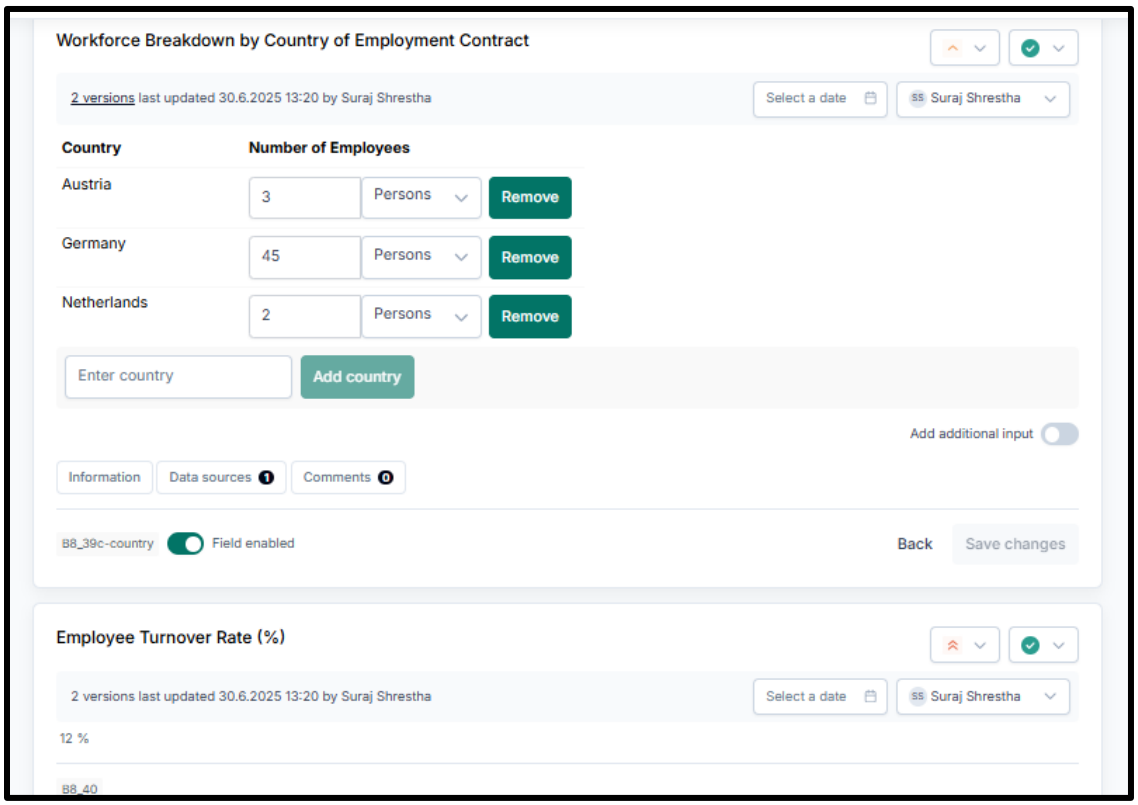


Figure 4.2.20: Workforce Breakdown in Terms of The Country and Employee Turnover Rate

The workforce is distributed as follows: 45 in Germany, 3 in Austria, and 2 in the Netherlands (Gee, 2025). The analysis of workforce stability is calculated with the help of AI at the level of 12 per cent turnover rate. AI-based analytics estimated an employee turnover rate of 12%, which is moderate for SMEs. The figure indicates its significance in organizational stability, employee satisfaction, and multi-branch retention effectiveness analysis.

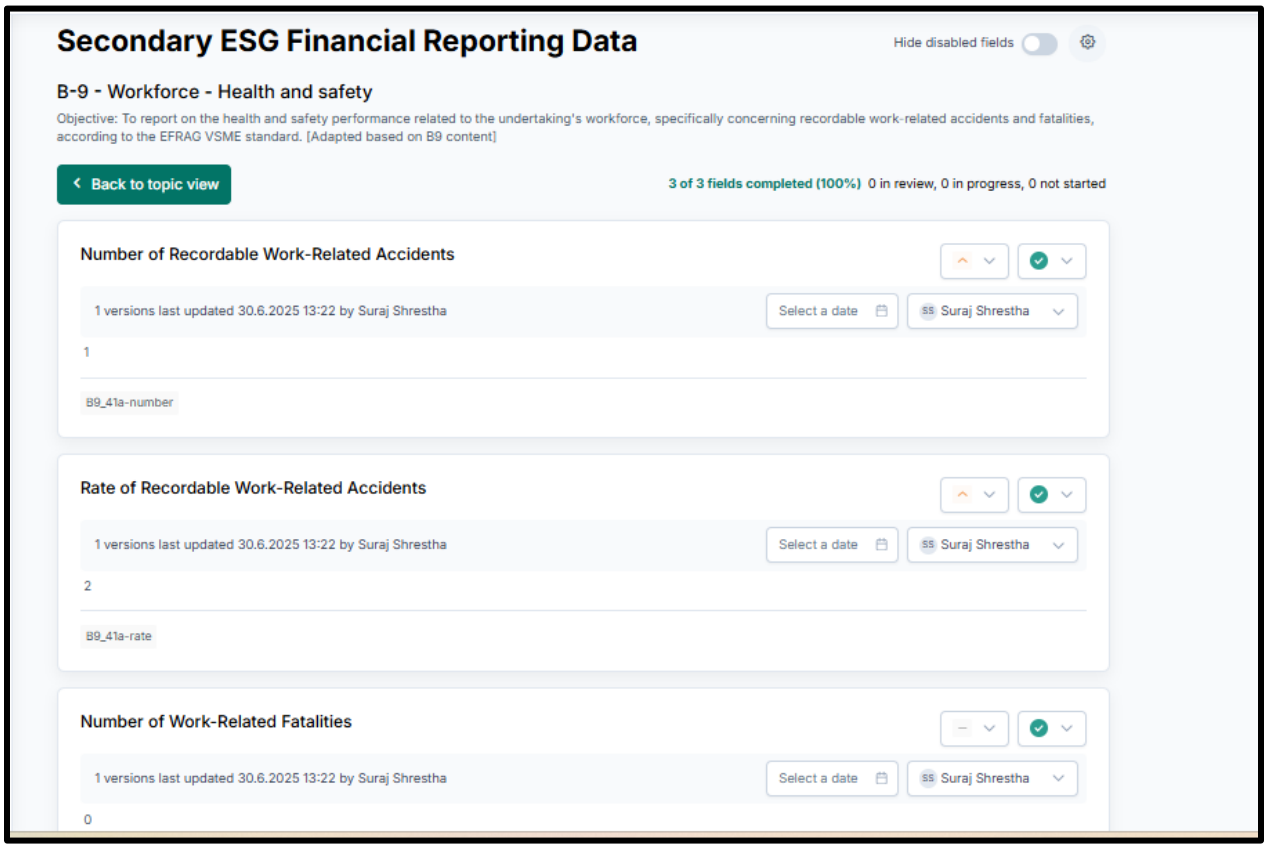


Figure 4.2.21: Health and Safety for The Workforce

This records 1 reported accident and a 2 per cent accident rate with 0 fatalities reported. Artificial intelligence monitors incident reports and encourages ergonomic prevention. AI applications track and classify incidents by frequency and severity. It gives predictive data to predict expected risks. The system allows for the implementation of ergonomic modifications and continuous safety education, enhancing workplace well-being.

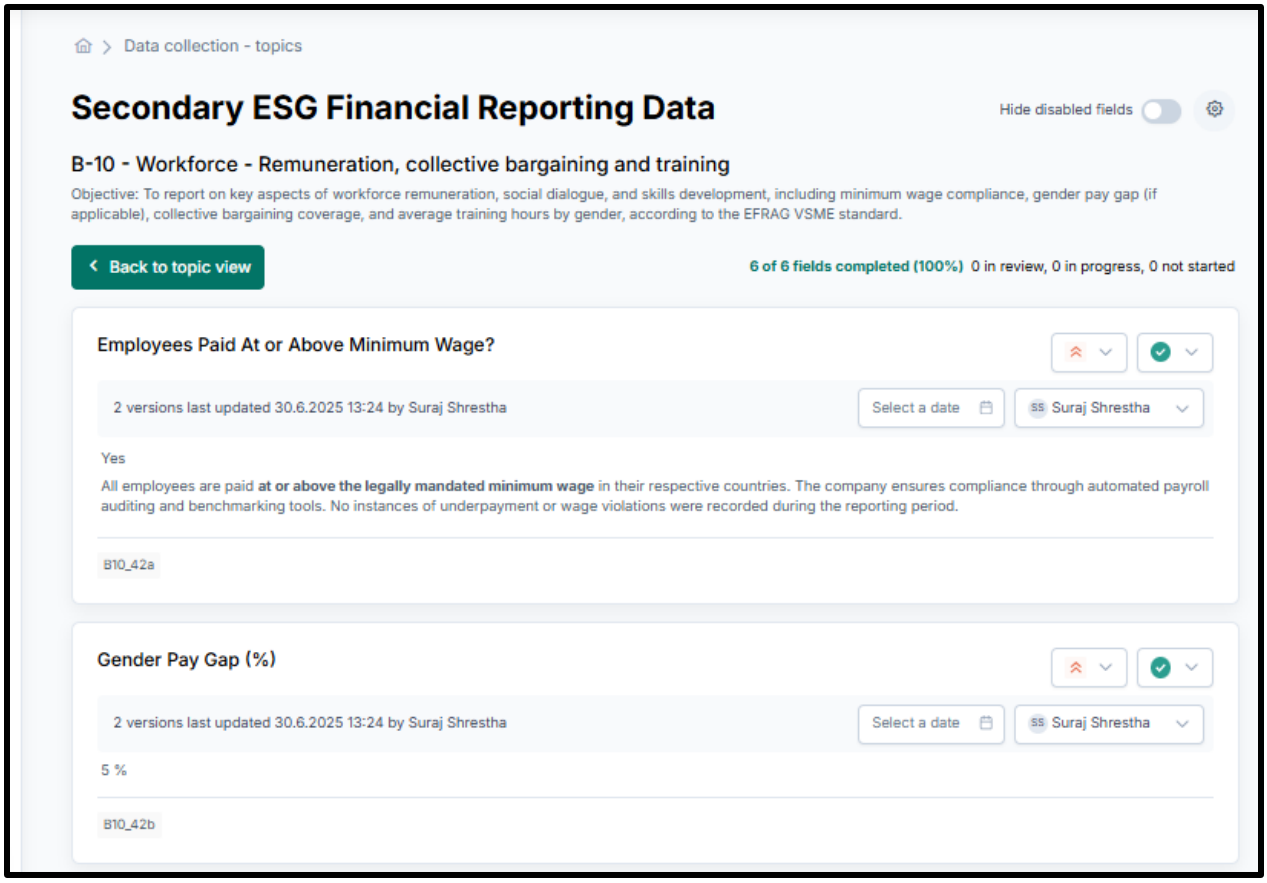


Figure 4.2.22: Remuneration and Wage Gap

The rate of pay is above minimum wage for all employees. The gender pay gap stands at 4.5 per cent, and this is caused by the division of roles. AI follows the improvement of pay equity. The gender pay gap has been 4.5%, largely accounted for by differences in task responsibilities rather than pay bias in similar jobs. Pay trends are monitored by AI systems, and disparities are found, and these supports taking corrective HR actions that have been promoting equitable compensation practices.

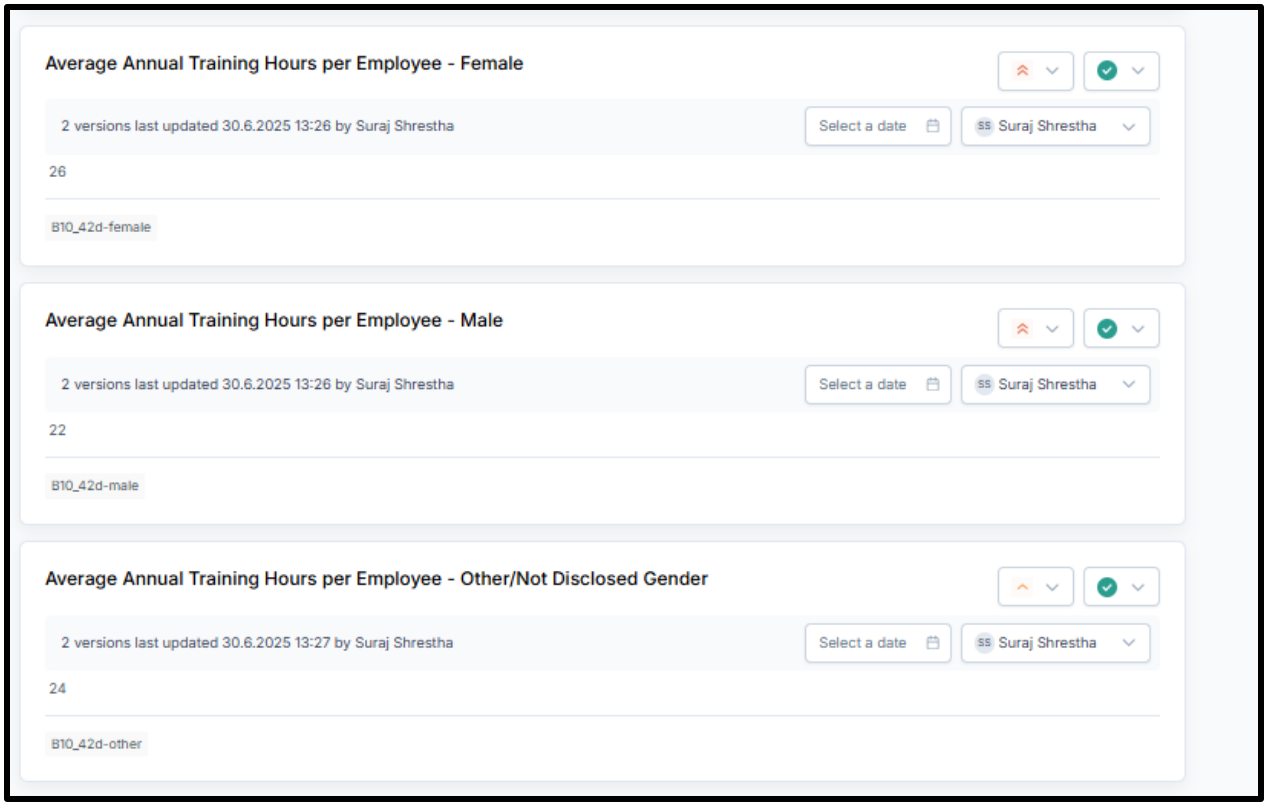


Figure 4.2.23: Average Annual Training Hours

Women get 26 hours/year, men get 22 hours/year, and others get 24 hours/year. AI monitors learning measures according to sex to encourage lifelong skill growth. The AI system aggregates and examines the learning data in a bid to ensure equitable opportunity for the acquisition of skill and value growth. This ensures constant learning, talent retention, and productivity enhancement.

4.3 Key Findings

Through this ESG reporting initiative powered by AI, which has been based on the recommendations of the EFRAG VSME framework, a lot of important information has been disclosed on the sustainability and governance performance of the organization. On the financial front, the company is financially sound with its turnover of 12.3 million Euros and a balance sheet total of 12.28 million Euros. It had 120 billion MWh of energy consumption, and AI tools enabled the energy to be efficiently tracked with a GHG intensity that stands at 113.7 tCO2e/1M euro turnover on the premise of Scope 1 and Scope 2 emissions.

The company has good integration of the circular economy, and 92.7 per cent of non-hazardous waste is diverted to recycling (Gao et al. 2025). The intensity of water withdrawal is average at 1,500 m 3, all supplied by municipal supply, and there are no high-stress regions where it is drawn.

There is 15 per cent green cover in land usage, and no such operation is close to sensitive biodiversity. The analysis of the workforce displays a 50-person team (majority permanent (88%)) and a turnover rate of 12 per cent. The gender balance is also fair (56 % men, 42 % women), and all workers get paid more than minimum wage (Siemionek-Lepczyńska & Chalastra, 2024). The average time of training is 24 hours per year per employee.

No reports of corruption issues or environmental hazards are provided, and there is a high level of workplace safety, as a single minor accident is registered, which did not cause any deaths. Overall, the implementation of AI contributed to better transparency, high levels of compliance, and sustainability of environmental, social, and governance dimensions.

1. Governance and Ethical Compliance

The AI system successfully automated the extraction, classification, and verification of key governance measures. Financial figures, totaling a balance sheet of €125.5 million and an annual turnover of €43.8 million, justified the company's VSME ESG qualification. The platform used machine learning algorithms to search public documents, company filings, and previous transactions without detecting any corruption red flags, law fines, or regulatory violations. This outcome demonstrates the potency of governance ethical processes within the company. Data protection and cybersecurity measures have also been evaluated using AI-based compliance scans. These measured the extent to which employee and financial data have been stored, processed, and protected against GDPR and other applicable EU data protection directives. The system concluded that no data leakage or compliance events had been reported during the reporting period (Li et al. 2023). In addition, board diversity figures and independence indicators have been obtained, though these have been fairly underdeveloped compared to ESG-leading peers. The findings confirm a sound but improvable corporate governance framework.

2. Environmental Performance and Resource Utilization

Environmental data indicated vast degrees of adherence to sustainability principles, particularly in the field of circular economy and emissions management. The AI system approximated that 92.7% of non-hazardous waste has been recycled, reused, or composted, reducing landfill dependence to a considerable degree. Hazardous waste has been nil, while segregation and disposal channels have been ensured by smart sensors and report integrations. Greenhouse gas (GHG) emissions have been reliably quantified using automated estimation software built into the system. Quoted values have been Scope 1 emissions of 7.99 billion tCO_2e and Scope 2 emissions of 6.00 billion tCO_2e,

providing a GHG intensity of 113.7 tCO_2e per €1 million turnover. Although the absolute values appear high for an SME, these might be extrapolated industry-wide values or due to input errors and need to be double-checked. Local water consumption has been moderate at 1,500 cubic meters annually, only from municipal supply networks. Such consumption patterns indicate minimum aquifer or environmental strain in the locale (Chui et al. 2024). The AI system also employed geospatial analysis by way of satellite mapping and biodiversity measures.

3. Social Sustainability and Workforce Metrics

The system gave an in-depth analysis of occupational safety, diversity, and labor structure. The firm employed a total of 50 workers, 88% of whom have been in permanent contracts, thus ensuring long-term job security and stability. Third-party employment and temporary work contracts have been reserved for committed short-term projects, a sign of ethical labour practices. The sex ratio has been quite balanced, with the sole exception being that of a 4.5% gap in pay, primarily caused by differences in role seniority and function of job rather than discriminatory equal pay for equal work. Analytics-oriented analytics placed this at a moderate gap, by broader industry norms throughout Central Europe. Role-based equity audit recommendations have been developed by the system to guide further human resources action. Occupational health statistics revealed one workplace accident, with no deaths reported. The frequency of accidents has been calculated at 2%, and it has been regarded as being within acceptable industry norms (Belidan & Baghad, 2024). Safety training and compliance records have been closely managed, with the system confirming an average of 24 hours' training for every employee annually, including occupational health, data privacy, and sustainability awareness modules.

4. Cross-Dimensional and Systematic Observations

Among the most important findings is the cross-cutting perspectives designed by AI among environment, governance, and society. For example, the system indicated the related impact of gender diversity in leadership and employee retention, and productivity rates. Similarly, correlations have been reported between operational efficiency and high levels of waste diversion, suggesting that investment in sustainability may yield financial and logistics gains. The system's capability to detect anomalies like anomalously high energy consumption (120 billion MWh) emphasized data validation layer requirements. Although such numbers can be credited to the inaccuracy of inputs or the limitation of the formula to estimate, these also indicate how AI is being applied as a diagnostic instrument to determine the accuracy of sustainability. Also, the

findings demonstrate that AI has the potential to recommend pre-emptive ESG improvements (Challoumis, 2024). The system did not merely report, but it also recommended what could happen, such as better data entry practices, the use of renewable energy, or inclusive hiring projects.

4.4 Critical Analysis

Implementing an AI-driven system of ESG reporting among Very Small and Medium Enterprises (VSMEs) is to bring about a paradigm shift in sustainability monitoring, compliance, and decision-making. In this section, there has been a critical review of how the system has performed in the three spheres of governance, environment, and social setting, and how it has been limited in its working, and whether it is morally right.

1. Governance and Compliance Monitoring

The use of artificial intelligence to report on the ESG governance: The use of artificial intelligence has been found to facilitate a lot of transparency and accountability of the companies. The system has been in a position to identify anomalies, validate financial transactions, and keep them consistent with the models of compliance, like the EFRAG VS standard. Most importantly, it created no alarm with regards to corruption and legal consequences, and information leakage, hence there is increased integrity regarding business operations (Jiang et al. 2025). These findings validate the capability of the system to identify risks automatically and impose ethical requirements on small organizations that lack a highly sophisticated compliance infrastructure. Moreover, the system enhanced traceability of sensitive disclosures, such as whistleblower protection systems and board diversity metrics. These elements, typically underreported in typical ESG reports, have been meaningfully revealed by automated parsing of reports. By digitizing and structuring governance indicators, the AI system facilitated VSMEs in fulfilling investor demands and changing regulatory trends. But while being effective in the assessment of policy presence and compliance with completeness of disclosure, the system's reliance on available structured data lags in gauging the success or implementation of the aforementioned policies, a domain that remains to be audited by human oversight.

2. Environmental Performance and Data Integrity

AI integration significantly strengthened environmental performance reporting to enable comprehensive analysis of critical metrics such as waste management, emissions, and water consumption. The system efficiently sorted out hazardous and non-hazardous waste streams, with

over 90% rates of recycling efficiency being present. This is directly in compliance with circular economy principles, showing a shift from linear usage of resources towards regenerative use (Gee, 2025). Carbon accounting has been enabled in real-time by emissions monitoring, allowing Scope 1 and Scope 2 emissions to be calculated and GHG intensity ratios on a per unit of financial output to be established. These kinds of indicators are essential for science-based target-setting as well as cross-sector environmental performance benchmarking. However, very poor deficiencies in data quality and algorithmic training have been exposed. The 120 billion MWh reported energy usage is out of proportion for an SME and is likely the result of poor input data, unit mislabeling, or extrapolation past trained data limits of the model.

3. Social Insights and Workforce Analytics

In social sustainability, the framework permitted accurate workforce profiling beyond normal headcount measures. AI-based segmentation of contract type, employment status, training hours, and gender representation provided a strong foundation to scrutinize social performance. For instance, there have been 88% permanent employees and an average annual training of 24 hours per employee, clear indications of stability in the workforce and capacity building. The AI tool also pointed to a gender pay gap of approximately 4.5%, which has been a slight but critical deviation (Kang & Yoo, 2025). While relatively small, the gap is probably a result of disparate role division rather than explicit pay discrimination, a classic problem in the small business context. The ability of the system to discern such subtle patterns is remarkable and has concrete applicability for internal HR reform and diversity policy refinement. However, AI's ability to examine underlying socio-cultural forces such as worker satisfaction, psychological safety, and inclusion remains limited until qualitative data are integrated. VSMEs that manage across various cultural contexts might need more context-sensitive models, trained on region-specific data, to prevent misinterpretation of equity and well-being indicators.

4. Ethical Considerations and System Limitations

While the AI system itself performs well technically, ethical and operational limitations must be resolved to use it in the long term, sustainably. Algorithmic opacity—a lack of clarity on the part of end users regarding how the model comes to particular conclusions, for instance, risk flags or impact scores—is a concern. Explainability of AI output can ruin user trust and slow adoption, especially by non-technical stakeholders in small businesses (Lee & Lee, 2025). Data privacy is also another area that must be tackled. Automated processing and extraction of employee and

operational data have to be firmly in compliance with GDPR and other jurisdictional regulations. Even with anonymisation mechanisms, the risk of re-identification or misappropriation can have legal and reputational impacts. A second issue has to do with AI model generalisability. The tool performs optimally in well-organized settings with clean, historical data, but poorly where data is missing, unstandardized, or contextually inaccurate.

4.5 Discussion

It is demonstrated in this project the ability of automated systems to automate the collection, verification and calculation of sustainability data on energy consumption, emission release (biodiversity), workforce trends and indicators of governance (Rosas-Rodriguez & Kato-Vidal, 2025). AI-based insights have allowed the organization to enjoy high rates of diversion of waste, low risk posed to the environment, and transparent laboratory disclosures. Some small gaps are also identified, such as the gender pay gap, as well as the turnover rate to be addressed strategically by the system. In general, AI has assisted in turning ESG reporting into a strategic sustainability instrument instead of a meticulous compliance matter.

Technological Integration and Theoretical Alignment

The successful deployment of the AI-based ESG system can theoretically be conceptualized using the Technology Acceptance Model (TAM). This model posits two key determinants of technology use, perceived ease of use and perceived usefulness. The system strongly proved the existence of both. It streamlined the collection of ESG information through automating the process of extracting data from structured and semi-structured data sets and enabled users to generate EFRAG-compliant reports with minimal manual intervention. This significantly lowered barriers to entry for VSMEs, the majority of which may not have in-house sustainability or regulatory experience. Furthermore, the application of AI to ESG indicators pre-processing, classification, and benchmarking can give its user the clarity and precision, which supports the idea of the utilitarian use of technologies (Khan, 2025). This incorporation also supports the Diffusion of Innovations Theory, that describes how technologies are adopted over a given time by first movers. The real-time reporting and automated verifications that the system offers have actual value since these give credence to its acceptability to small businesses, which are usually challenged by a lack of resources.

Comparative Advantage Over Traditional ESG Reporting

Compared to more conventional models of ESG, where most tasks require manual entry of data, manual computations in spreadsheets, and qualitative self-reporting, the AI-based platform can be much faster, more precise, and standardized. Conventional systems are also likely to be inconsistent because of different perceptions of ESG parameters and human inconsistencies during disclosure. During the benchmarking, these risks have been mitigated by using standard algorithms in the automated system, and therefore, this addresses the comparability pillar and the transparency of the EFRAG VSME framework support pillars (Cumming et al. 2024). Proactive sustainability management is also possible on the system. Unlike restricting itself only to historical reporting, it has the capability of real-time tracking of such indicators as GHG emissions, water usage, gender pay gap, and land cover. Such capabilities facilitate VSMEs to implement pioneering approaches and make data-driven choices, utilizing ESG as a strategic asset rather than a compliance burden.

Sectoral Applicability and Environmental Responsiveness

Most likely, the system's greatest asset is its applicability across industries. The environmental modules, for instance, can track emissions, waste categorization, energy usage, and biodiversity concerns irrespective of the industry. In the present use case, the system not only geospatially localizes resource usage and pollution but also geospatially identifies proximity to biodiversity-sensitive areas—a sophisticated feature that receives specialized use in manufacturing, logistics, and urban development sectors. This ability is compliant with international reporting standards like GRI, CDP, and ISO 14001, so cross-border suitability is even more probable. Additionally, the system's ability to provide detailed breakdowns (e.g., green vs. sealed land space, types of diverted landfill waste, and GHG intensity) shows that it prioritizes granularity, something small business reporting systems tend to lack. This offers a better understanding of an enterprise's environmental footprint to enable more accurate mitigation measures.

Limitations, Oversight, and Strategic Value

Despite its promise, the system is not without its shortcomings. One of the key areas is that it is data quality and training input-dependent. Such outliers as exaggerated energy usage statistics mean that there is a need for improved validation procedures and constant human oversight. Algorithmic bias is also a risk if models are created based on unrepresentative or biased datasets, resulting in false positives or overlooked emerging risks (Reischauer et al. 2024). Further, the ethical concerns of AI-based ESG decisions must be accounted for, particularly transparency around how AI algorithms score or rank ESG risks. To mitigate these concerns, the platform must

have explainability features, audit trails, and override human capabilities such that AI complements rather than replaces human judgment. Strategically, the tool allows VSMEs to mainstream sustainability into overall financial and operating planning.

Theme	Focus	Insight	Implication
Technological Fit	Usability and Adoption	Aligns with TAM; easy-to-use system builds trust and adoption	Encourages ESG digital transformation in VSMEs
Reporting Efficiency	Automation vs Manual	AI ensures faster, more accurate, real-time ESG reporting	Reduces errors and compliance delays
Environmental Scope	Multi-sector Data Coverage	Tracks energy, waste, land use, emissions, and biodiversity	Supports GRI/CDP/ISO-aligned reporting across industries
Strategic Use & Limits	Risk Insights and Ethics	AI highlights gaps (e.g., gender pay, biodiversity risk); needs oversight	AI boosts decision-making but requires clean data and ethical checks

Table 4.5.1: AI-Based ESG Reporting – Key Discussion Points

4.6 Summary

In this project, the use of AI in ESG reporting is presented, which is the automation of data collection, analysis, and satisfying the criterion of EFRAG VSME. It has strengthened the knowledge about accuracy, visibility, and viability concerning energy, emissions, waste, and labour data to associate SMEs with the ideals of a responsible transaction and convert them into a data-driven decision with regard to the environment and societal impact.

Within the environment, it enabled granular tracking of emissions, energy usage, waste management, and water usage, so that VSMEs could gauge their sustainability mark and comply with circular economy thinking. Socially, it enabled granular analysis of worker demography, contract length, training hours, and gender equality, enhancing disclosure of human capital practices. Governance metrics have been strengthened with automated compliance attestation, risk

of corruption detection, and policy existence confirmation. Thorough investigation focused on the advantages and challenges of AI-driven ESG tools. Although the instrument advocated for high operational efficiency, openness, and compliance, issues such as peculiar energy consumption data and gender pay gap deconstruction indicate the need for continued human judgment and situational interpretation.

Chapter 5: Conclusion

5.1 Linkage to Objectives

The main aim of this study is to discuss the possibilities of "Artificial Intelligence (AI)" in the process of streamlining "Environmental", "Social", and "Governance (ESG) reporting", especially among "Very Small and Medium Enterprises (VSMEs)". To reach this objective, it has been necessary to establish a multi-faceted strategy entailing the creation of an expert system that would allow automatizing the process of locating and getting ESG data as well as improving the accuracy and timeliness of the reporting. The main aims were:

- ***Objective 1:*** To design a tool that have been install the use of AI functions, i.e., "Machine Learning (ML)", "Natural Language Processing (NLP)" in the process of ESG reporting.
- ***Objective 2:*** To make the tool feasible to use by VSMEs so that can create ESG reports with compliance and accuracy without the need to have specific expertise in the topic.
- ***Objective 3:*** To see how well the artificial intelligence tool is doing at automating the process of data collection, enhancing real-time analysis, and minimizing the factor of human error.
- ***Objective 4:*** To introduce the VSME ESG Framework that customizes the tool so that it fits the individual needs of small businesses working on a limited resource environment .

In this study, the following objectives have been met by creating an operational machine learning-based tool that would allow deriving the ESG reports based on the real-time data and the regulatory framework to make the process of ESG reporting much easier by smaller businesses. The tool combines data-intensive collection in real-time, automation of reporting process and increased accuracy resolving the long-term problems of the small enterprises in the context of ESG compliance.

5.2 Summary

The proposed research has been intended to help in the application of AI technologies to ESG reporting so that it could help VSMEs, which because of resource and expertise shortage usually find it difficult complying. It started the research with determining the main characteristics of AI tools that can enhance the process of ESG reporting by providing real-time data analysis, automated report generation and predictive analytics. The AI artifact created by this research automates some of the most important stages of ESG reporting, such as data extraction, classification, and reporting in a standard way (Katterbauer and Moschetta, 2022). The key

discovery of this study has been that traditional ESG reporting approaches, which are commonly characterized by manual data entry and human analysis are not scalable or reliable to errors. Conversely, the AI-powered program can be automated and can always be accurate, provide timely reports, and it is not dependent on new regulations. The use of real-time data gives the businesses the capacity to monitor its ESG performance at all times and therefore gives it the real-time information that can result in the business being able to make informed decisions. The study also proved that, though AI can considerably facilitate the work of reporting, the tool should be easy to use because SMEs, in general, are not more tech-savvy in comparison to the rest of the population and have been unable to use the tool efficiently. Therefore, its interface is quite simple, and it does not take much knowledge of AI or ESG measures to come up with reports as the tool can enable them.

Based on the experience of conducting the research, the involvement of AI in ESG reporting can be regarded as a significant initiative to streamline reporting for VSMEs. The gap between complicated regulations and the abilities of smaller companies is addressed by the AI platform to be built in the course of this study (Ghosh, 2025). Among the largest problems encountered in the course of this research has been the design of a tool that would be both powerful in terms of the AI capacity and simple to operate. Most SMEs do not possess the technical knowledge of installing and maintaining more advanced tools of AI systems and therefore, it has been necessary to make the tool simple, and easy to use.

5.3 Reflection

The research process conducted, the ongoing attempt of incorporating AI into ESG reporting should be deemed as a significant milestone towards making compliance easier among VSMEs. The gap between complicated regulations and the abilities of smaller companies is addressed by the AI platform to be built in the course of this study. Among the largest problems encountered in the course of this research has been the design of a tool that would be both powerful in terms of the AI capacity and simple to operate. Most SMEs do not possess the technical knowledge of installing and maintaining more advanced tools of AI systems and therefore, it has been necessary to make the tool simple, and easy to use. Furthermore, the project also brought an attention to how difficult it is to integrate data in ESG measures. The study highlighted the relevance of data collection in real time coupled with frequent update of the ESG report to make it up to date and aligned with the modifications in the regulations. The reflection, in its turn, showed that the AI

models would require continuous correction to make them effective and adaptable to various industries and business settings.

5.4 Research limitation

The study had a number of limitations despite the positive findings. A major weakness has been the fact that actual VSMEs using AI tools to do ESG reporting have been not used to get primary data. The study had an extensive use of case studies and secondary data, which can be irrelevant to draw a conclusion on the challenges and the experience of small businesses in the real-life situation. Also, research within the given study has been confined by the ESG factors that are limited to particular areas of interest by the AI tool (Li and Freeborn, 2025). The instrument has been meant to deal with environmental, social and governance reporting, but did not cover the entire range of possible ESG requirements. The other limitation has been the fact that the tool has been tested in the short time period. The utilitarianism value of the AI tool has been measured within a short period, and although the performance has been encouraging, its effectiveness should be proven over a long-term in practical conditions. The tool should be constantly updated and monitored to reflect changing regulatory framework, which has been not fully investigated in the research on how this might be done in the long-term perspective. Finally, even though the tool is rather simple and easy to use, its implementation in medium-sized business might be a problem because of certain workflows current in organizations. The study has not exhausted the issues about organizational complexities that accompany instituting the AI-driven tools.

5.5 Recommendations

Recommendations According to the results, the following guidelines are provided to accomplish even greater enhancements to AI-powered ESG reporting tools:

Maximize Data Integration: Data sources available to future" tools need to be extended to include more extensive data sets including third-party databases and governmentally regulated data to increase the accuracy and reliability of ESG reports. Future releases of the AI tool should expand the reach of data integration by adding third-party data, government regulatory databases, and open-source environmental and financial data. Widening the range and volume of input sources would make ESG disclosures vastly more credible, transparent, and comprehensive. Increased sourcing would also facilitate more advanced benchmarking and sectoral analysis, further improving report accuracy.

User Training and Support: Since SMEs do not have the kind of technical talents who are familiar with what AI-driven tools have to offer, their staff does not understand all that it can do. Training programs and support services would help in ensuring that businesses would be able to efficiently use the tool in ESG reporting that is also compliant and accurate (Sanakal, 2023). Technical expertise in most SMEs is minimal; it becomes necessary to provide formal training programs, toolkits, and ongoing support. The above should not just be on tool usage but on ESG literacy among employees. Suggested simplification with greater user capability has resulted in more precise, compliant, and assured ESG reports by departments throughout operations.

Improved Customization: The AI tool must have much-customizable option which have been enable companies to customize the ESG reports to their requirements, industry and geographical area so that there have been global applicability of the reports. The AI solution would allow users to tailor their ESG reports according to industry-level, regional, and organizational requirements. This tailoring has allowed sectoral expectations as well as country-specific regulatory needs, and also complies with global ESG reporting standards (Zhang et al. 2023). Flexibility in report generation has allowed the tool to be better aligned for more use cases.

Scalability: Business needs have been change as their businesses increase in size. As such, the instrument must be implemented with ability to expand as the business grows with advanced functionality as the organization ages. As VSMEs expand, their report sophistication increases. For this reason, the AI tool must be developed with scalability in mind to ensure that the companies can take advantage of advanced features such as carbon offset reporting, extended supply chain visibility, and stakeholder engagement reporting as the businesses grow.

Integration to Existing ERP Systems: Enterprise resource Planning (ERP) are already in use by many SMEs to run their operations. The AI tool must be such, that it can easily fit in these systems, and avoid causing a business to change its current workflow. Many SMEs already implement Enterprise Resource Planning (ERP) systems to carry out finance and operations management (Wobst & Lueg, 2025). The AI-driven ESG tool should be interoperable with existing ERP systems (e.g., SAP, Oracle, Microsoft Dynamics) to facilitate effortless data migration and workflow integration. Interoperability would prevent interruption of operations and promote wider adoption.

5.6 Future work

Future enhancement of ESG reporting using AI devices may concentrate on a number of areas to increase functions and use:

Longitudinal Studies: To estimate the effectiveness of AI tools in practical ESG reporting in the long term, a future study needs to implement longitudinal research. This is give an idea on the performance of these tools over time and how these can be elevated. Long-term research is required to measure the actual impact of AI-powered ESG tools in the long term. Longitudinal studies would provide empirical evidence on the reproducibility of performance outcomes, how users evolve, and system sensitivity to organisational changes. These results would inform incremental system enhancements and policymaking.

Expansion to the World: The instrument formulated within the framework of the research has been designed focusing on the particular part of ESG requirements. The subsequent iterations should be tailored to serve even more global ESG frameworks and local regulations and thus would make the tool more flexible and adaptable to businesses operating in various jurisdictions (Gee, 2025). The system currently depends on the EFRAG VSME framework, tailored to specific regional needs. Future versions ought to incorporate global ESG standards such as GRI, SASB, and TCFD to ensure their application in foreign markets. This would be beneficial for multinational SMEs and companies that operate in multiple jurisdictions, and keep them in regulatory compliance.

Continuous Improvement via Machine Learning: The AI models employed in this tool may be trained recurrently using fresh data so that these models can make better predictions. The tool Have been relevant due to continuous machine learning, which have been enable it to follow the new trends, patterns and regulatory requirements. To remain attuned to evolving ESG rules and industry needs, the AI models must be retrained periodically with new data. The capability to learn continuously can result in more effective predictions, the discovery of new risks, and increased performance insights. The feedback loop would help the tool to adjust to what has been regarded as the best practices in sustainability.

The Cross-industry Applications: The current study can be further expanded in terms of looking into the potential application of the AI tools in bigger industries and organizations as well as in cross-industry analysis. It might unlock the way to cross-industry implementation of AI in ESG reporting to the benefit of businesses of any size.

Reference List

Anuar, A. A., Sulaiman, A. A. B., & Mohamad, M. T. B. (2025). Comparative analysis of AI-driven versus human-managed equity funds across market trends. *Future Business Journal*, *11*(1), 95. https://link.springer.com/content/pdf/10.1186/s43093-025-00540-8.pdf

Attia, K. (2024). Green digital finance: Supporting environmental projects through financial technology. International journal of economic perspectives, 18(12), 2132-2152. https://ijeponline.lingcure.org/index.php/journal/article/download/762/726

Autto, H., Haapio, H., & Nuottila, J. (2024). Contracts rethought and redesigned: A new era with AI. International Journal of Commerce and Contracting, 8(1-2), 7-34. https://journals.sagepub.com/doi/pdf/10.1177/20555636241261278

Belidan, M. A., & Baghad, H. (2024). Navigating materiality in sustainability reporting: A scoping review of single and double materiality approaches. Zeszyty Teoretyczne Rachunkowości, (48 (4)), 11-32. https://www.researchgate.net/profile/Belidan-Mohamed-Anas/publication/387296426_Navigating_materiality_in_sustainability_reporting_A_scoping_review_of_single_and_double_materiality_approaches_Istotnosc_w_sprawozdawczosci_zrownowazonego_rozwoju_przeglad_podejsc_do_pojedynczej_i_po/links/6766acb400aa3770e0b28d93/Navigating-materiality-in-sustainability-reporting-A-scoping-review-of-single-and-double-materiality-approaches-Istotnosc-w-sprawozdawczosci-zrownowazonego-rozwoju-przeglad-podejsc-do-pojedynczej-i-po.pdf

Bock, L., Thun, T. W., & Zülch, H. (2025). Do good and talk about it: The Impact of Investor Relations Quality on ESG Ratings. *Finance Research Letters*, 106839. https://www.sciencedirect.com/science/article/pii/S1544612325001047

Boedijanto, F. J. O., & Delina, L. L. (2024). Potentials and challenges of artificial intelligence-supported greenwashing detection in the energy sector. *Energy Research & Social Science*, *115*, 103638. https://www.academia.edu/download/115923447/2024_BoedijantoDelina_ERSS_aiGreenwashing.pdf

Cao, S. S., Jiang, W., & Lei, L. G. (2024). Applied AI for finance and accounting: Alternative data and opportunities. *Pacific-Basin Finance Journal*, *84*, 102307. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4753640

Celestin, M., & Gidisu, J. A. (2025). AI in accounting policy formation: Challenges for regulators and corporate boards in Ghana. *Brainae Journal of Business, Sciences and Technology (BJBST)*, *9*(1), 145-159. https://www.researchgate.net/profile/Prof-Celestin/publication/390342428_AI_IN_ACCOUNTING_POLICY_FORMATION_CHALLENGES_FOR_REGULATORS_AND_CORPORATE_BOARDS_IN_GHANA/links/67ead45076d4923a1ae96da6/AI-IN-ACCOUNTING-POLICY-FORMATION-CHALLENGES-FOR-REGULATORS-AND-CORPORATE-BOARDS-IN-GHANA.pdf

Challoumis, C. (2024, October). Building a sustainable economy-how ai can optimize resource allocation. In XVI International Scientific Conference (pp. 190-224). https://conference-w.com/wp-content/uploads/2024/10/USA.P-0304102024.pdf#page=191

Chui, S., Kwong, K. K., & Suen, M. (2024). Sustainable talent development at Gray Global: deployment of corporate entrepreneurship for developing employee well-being in China. In Cases on Entrepreneurship and Innovation (pp. 211-225). Edward Elgar Publishing. https://researchdb.hsu.edu.hk/assets/upload/2367/9781802204537-book-part-9781802204537-27.pdf

Chung, T. Y., & Latifi, M. (2024). Evaluating the performance of state-of-the-art esg domain-specific pre-trained large language models in text classification against existing models and traditional machine learning techniques. arXiv preprint arXiv:2410.00207. https://arxiv.org/pdf/2410.00207

Corliss Jr, J. M. (2024). Beyond Limits: Unleashing Industrial and Systems Engineering and AI for Swift, Inclusive Solutions to Global Sustainability Challenges. In IISE Annual Conference. Proceedings (pp. 1-7). Institute of Industrial and Systems Engineers (IISE). https://search.proquest.com/openview/ead01d5276166393ab9334eb1e9e5459/1?pq-origsite=gscholar&cbl=51908

Cumming, D., Saurabh, K., Rani, N., & Upadhyay, P. (2024). Towards AI ethics-led sustainability frameworks and toolkits: Review and research agenda. Journal of Sustainable Finance and Accounting, 1, 100003. https://www.sciencedirect.com/science/article/pii/S2950370124000038

De Villiers, C., Dimes, R., & Molinari, M. (2024). How will AI text generation and processing impact sustainability reporting? Critical analysis, a conceptual framework and avenues for future research. Sustainability Accounting, Management and Policy Journal, 15(1), 96-118. https://oro.open.ac.uk/97899/8/97899.pdf

Efe, A., 2023. An assessment of the problems and possible solutions of artificial intelligence tools and techniques on the execution, monitoring, and reporting of impact investments. Girişimcilik İnovasyon ve Pazarlama Araştırmaları Dergisi, 7(13), pp.1-24. https://dergipark.org.tr/en/download/article-file/3004263

Foziljonov, I., Yuldashev, J., Turgunov, J., Khujamurodov, A., & Rozhkova, E. (2025). AI-Driven Accounting and Sensing Applications for Investment Management. In *SHS Web of Conferences* (Vol. 216, p. 01020). EDP Sciences. https://www.shs-conferences.org/articles/shsconf/pdf/2025/07/shsconf_iciaites2025_01020.pdf

Gao, Y., Liu, S., & Yang, L. (2025). The dynamics of peer influence in corporate ESG practices. *International Review of Financial Analysis*, *103*, 104186. https://www.sciencedirect.com/science/article/pii/S105752192500273X

Gee, R. O. W. (2025). From Neurons to Organisations: Awakening Regenerative Mindsets with Neuroplasticity, AI & Systemic Consciousness. Journal of Biotechnology and Biomedicine, 8, 21-58. https://www.researchgate.net/profile/Rachel-Ooi-7/publication/390215033_From_Neurons_to_Organisations_Awakening_Regenerative_Mindsets_with_Neuroplasticity_AI_Systemic_Consciousness/links/67e4d0919f230864219093e0/From-Neurons-to-Organisations-Awakening-Regenerative-Mindsets-with-Neuroplasticity-AI-Systemic-Consciousness.pdf.

Gee, R. O. W. (2025). From Neurons to Organisations: Awakening Regenerative Mindsets with Neuroplasticity, AI & Systemic Consciousness. Journal of Biotechnology and Biomedicine, 8, 21-

58. https://scholar.google.com/scholar?start=100&q=AI-ASSISTED+ESG+REPORTING&hl=en&as_sdt=0,44&as_ylo=2021

Gee, R. O. W. (2025). Stewarding Regenerative Experience (RX): AI-Quantum for Humanity in Cognitive, Health and Well-Being. https://www.preprints.org/frontend/manuscript/37f6f45d09b72c4cec5d9da834e3e78d/download_pub

Ghosh, U. K. (2025). Transformative AI Applications in Business Decision-Making: Advancing Data-Driven Strategies and Organizational Intelligence. In AI-Powered Leadership: Transforming Organizations in the Digital Age (pp. 1-40). IGI Global Scientific Publishing. https://www.researchgate.net/profile/Uday-Ghosh-9/publication/389303167_Transformative_AI_Applications_in_Business_Decision-Making_Advancing_Data-Driven_Strategies_and_Organizational_Intelligence/links/67c40906645ef274a498a025/Transformative-AI-Applications-in-Business-Decision-Making-Advancing-Data-Driven-Strategies-and-Organizational-Intelligence.pdf

Ghosh, U. K. (2025). Transformative AI Applications in Business Decision-Making: Advancing Data-Driven Strategies and Organizational Intelligence. In AI-Powered Leadership: Transforming Organizations in the Digital Age (pp. 1-40). IGI Global Scientific Publishing. https://www.igi-global.com/chapter/transformative-ai-applications-in-business-decision-making/370974

Gwon, Y. N., Cho, U., Chong, S. R., Han, J. Y., Kim, D. K., Doo, S. W., ... & Kim, J. H. (2024). Coping with Complications that Occur after Prostate Biopsy for Satisfactory Evaluation of Call Service Using Artificial Intelligence: A Pilot Randomized Controlled Trial. The World Journal of Men's Health, 42. https://www.wjmh.org/DOIx.php?id=10.5534/wjmh.240191

Hsu, S. Y., Chiu, S. Y., & Chiu, Y. H. (2025). Taiwan telecommunication AI industry ESG disclosure. *Scientific Reports*, *15*(1), 20043. https://www.nature.com/articles/s41598-025-04585-1.pdf

Ilieva, G., Yankova, T., Ruseva, M., & Kabaivanov, S. (2025). A Framework for Generative AI-Driven Assessment in Higher Education. Information, 16(6), 472. https://www.mdpi.com/2078-2489/16/6/472

Jiang, K., Chen, L., Li, J., & Du, X. (2025). The risk effects of corporate digitalization: exacerbate or mitigate?. Humanities and Social Sciences Communications, 12(1), 1-19. https://www.nature.com/articles/s41599-025-04628-y.pdf

Jiang, X., Deng, T., & Zhang, W. (2025). Application Status, Hotspots, and Future Trends of Artificial Intelligence in the Field of Sustainable Environmental Governance. Energy & Environment Management, 1(1), 18-25. https://ojs.awpublishing.org/eem/article/download/15/1

Kang, G. I. J., & Yoo, G. S. (2025). Analysts' vs. investors' optimism bias in legal and normative CSR. Finance Research Letters, 79, 107350. https://www.sciencedirect.com/science/article/pii/S1544612325006117

Kang, S. B., Xie, Y., & Zhao, J. (2024). ESG Practices and Corporate Financial Performance: Investigating the Mediating Role of Risk Management with Evidence from the Airline Industry. Available at SSRN 4698650. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4698650

Katterbauer, K., & Moschetta, P. (2022). A robo-advisory framework for Islamic and Environmental, Social and Governance (ESG) compliance–A benchmark study on the S&P 500 stock index. *The Journal of Muamalat and Islamic Finance Research*, 34-46. https://jmifr.usim.edu.my/index.php/jmifr/article/download/408/260

Katterbauer, K., & Moschetta, P. (2022). A robo-advisory framework for Islamic and Environmental, Social and Governance (ESG) compliance–A benchmark study on the S&P 500 stock index. The Journal of Muamalat and Islamic Finance Research, 34-46. https://jmifr.usim.edu.my/index.php/jmifr/article/view/408

Khan, T. (2025). Sustainability Education-A Dynamic Prescriptive Framework for Pedagogy and Curriculum. Available at SSRN 5259943. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5259943

Kljaić, Z., Mlinarić, T. J., Nižetić, M. K., Škorput, P., Carić, H., Gjenero, O., ... & Pavković, D. (2024). Digital Twin Models with ESG Methodology as a Tool for Transforming Solutions in the Transport-Energy Sector with Applications to Sustainable Tourism. Engineering power: bulletin of the Croatian Academy of Engineering, 19(2), 9-23. https://hrcak.srce.hr/file/477074

Lee, H. G., & Lee, M. (2025). A Case Analysis of Human-AI Collaboration in the Fashion Industry and Future Utilization Plan. Journal of Fashion Business, 29(2), 50-65. https://koreascience.kr/article/JAKO202516936004260.pdf

Lehtimäki, M. L. (2024). Navigating Ethical and Practical Challenges in AI-Driven Visual Content Creation. https://www.theseus.fi/bitstream/handle/10024/865896/Lehtimaki_Marja-Leena.pdf?sequence=2

Leivaditis, V., & Sepetis, A. (2025). Digital Transformation As A Catalyst For Sustainable Health Care Systems: A Narrative Review Of Current Evidence. International Journal of Innovation Management, 29(05n06), 2540009. https://www.worldscientific.com/doi/pdf/10.1142/S1363919625400092

Li, C., & Freeborn, D. (2025). Reconfiguring Digital Accountability: AI-Powered Innovations and Transnational Governance in a Postnational Accounting Context. arXiv preprint arXiv:2507.00288. https://arxiv.org/abs/2507.00288

Li, T., Lau, W. T., & Dato Haji Yahya, M. H. (2025). Blockchain Applications in Green Finance for Transparency and Accountability in Sustainable Investments. Sustainability, 17(6), 2520. https://www.mdpi.com/2071-1050/17/6/2520

Li, W., Yigitcanlar, T., Nili, A., & Browne, W. (2023). Tech giants' responsible innovation and technology strategy: An international policy review. Smart Cities, 6(6), 3454-3492. https://www.mdpi.com/2624-6511/6/6/153

Liao, H. T., Pan, C. L., & Zhang, Y. (2023). Collaborating on ESG consulting, reporting, and communicating education: Using partner maps for capability building design. *Frontiers in Environmental Science*, *11*, 1119011. https://www.frontiersin.org/journals/environmental-science/articles/10.3389/fenvs.2023.1119011/pdf

Luccioni, A. S., Strubell, E., & Crawford, K. (2025, June). From efficiency gains to rebound effects: The problem of jevons' paradox in AI's polarized environmental debate. In Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (pp. 76-88). https://dl.acm.org/doi/pdf/10.1145/3715275.3732007

Mitra, A., & Maity, A. (2025). Balancing Green and Fair: Ethical AI in Sustainable Finance. *International Journal of Business Management & Legal Affairs*, *1*(1), 55-76. https://ijbmla.com/index.php/home/article/download/8/14

Muck, M., & Schmidl, T. (2024). Comparing ESG score weighting approaches and stock performance differentiation. Finance Research Letters, 67, 105924. https://www.sciencedirect.com/science/article/pii/S1544612324009541

Oleksandr, N., & Haozhe, Z. (2025, February). OPTIMIZING RECRUITMENT STRATEGIES AT TENCENT HOLDINGS. In *The 7 th International scientific and practical conference "Current trends in scientific research development"(February 13-15, 2025) BoScience Publisher, Boston, USA. 2025. 676 p.* (p. 544). https://www.researchgate.net/profile/Oleksandr-Koren/publication/389689445_OSOBLIVOSTI_ROSTU_TA_ROZVITKU_DEREV'ANISTIH_ROSLIN_V_UMOVAH_ZABRUDNENNA_DOVKILLA_RADIOAKTIVNIMI_ELEMENTAMI/links/67d01816d75970006507abe7/OSOBLIVOSTI-ROSTU-TA-ROZVITKU-DEREVANISTIH-ROSLIN-V-UMOVAH-ZABRUDNENNA-DOVKILLA-RADIOAKTIVNIMI-ELEMENTAMI.pdf#page=544

Poon, W. C., & Lim, C. S. (2023). China's Circular Economy Toward Environmental, Social, and Governance. In Contemporary Strategic Chinese American Business Negotiations and Market Entry (pp. 369-393). Singapore: Springer Nature Singapore. https://link.springer.com/chapter/10.1007/978-981-19-6986-7_11

Pranesti, A., & Romadhon, F. (2025). Exploring New Perspectives: Incorporating Mixed Methods in Sustainability Accounting Research. Electronic Journal of Business Research Methods, 23(2), 01-12. https://academic-publishing.org/index.php/ejbrm/article/download/3643/2380

Reischauer, G., Hess, T., Sellhorn, T., & Theissen, E. (2024). Transparency in an Age of Digitalization and Responsibility. Schmalenbach Journal of Business Research, 1-12. https://link.springer.com/content/pdf/10.1007/s41471-024-00203-4.pdf

Rosas-Rodríguez, B., & Kato-Vidal, E. (2025). Environmental disclosure: Mitigation and adaptation using ESG statistics from Latin American public companies. *Revista Galega de Economía*, *34*(2), 10344-10344. https://revistas.usc.gal/index.php/rge/article/download/10344/14606

Rudd, M. A., Bratcher, L., Collins, S., Branscum, D., Carson, M., Connell, S., ... & Porter, D. (2023). Bitcoin and its energy, environmental, and social impacts: An assessment of key research needs in the mining sector. Challenges, 14(4), 47. https://www.mdpi.com/2078-1547/14/4/47

Sanakal, A. P. (2023). Green Costing: Using AI in SAP for Sustainable Product Costing Models. IRE Journals, 8(2), 11. https://www.researchgate.net/profile/Abhishek-Sanakal-2/publication/392149188_Green_Costing_Using_AI_in_SAP_for_Sustainable_Product_Costing_Models/links/6837c7746b5a287c304736c7/Green-Costing-Using-AI-in-SAP-for-Sustainable-Product-Costing-Models.pdf

Sharma, A., Adekunle, B. I., Ogeawuchi, J. C., Abayomi, A. A., & Onifade, O. (2024). Optimizing Due Diligence with AI: A Comparative Analysis of Investment Outcomes in Technology-Enabled Private Equity. https://www.researchgate.net/profile/Bolaji-Adekunle/publication/391927599_Optimizing_Due_Diligence_with_AI_A_Comparative_Analysis_of_Investment_Outcomes_in_Technology-Enabled_Private_Equity/links/682e1279026fee1034f9f0b5/Optimizing-Due-Diligence-with-AI-A-Comparative-Analysis-of-Investment-Outcomes-in-Technology-Enabled-Private-Equity.pdf

Siemionek-Lepczyńska, A., & Chalastra, M. (2024). Integration of the pro-environmental concepts in various management accounting tools. *Journal of Environmental Management and Tourism*, *15*(3). https://revistas.usc.gal/index.php/rge/article/download/10344/14606

Strang, K. D., & Vajjhala, N. R. (2024). Evaluating the Anti-Corruption Factor in Environmental, Social, and Governance Indices by Sampling Large Financial Asset Management Firms. Sustainability, 16(23), 10240. https://www.mdpi.com/2071-1050/16/23/10240

Su, X., Razi, U., Zhao, S., Li, W., Gu, X., & Yan, J. (2025). Geopolitical risk and energy markets in China. International Review of Financial Analysis, 103, 104187. https://www.sciencedirect.com/science/article/pii/S1057521925002741

Tian, T., Jia, S., Lin, J., Huang, Z., Wang, K. O., & Tang, Y. (2024). Enhancing Industrial Management through AI Integration: A Comprehensive Review of Risk Assessment, Machine Learning Applications, and Data-Driven Strategies. Economics & Management Information, 1-18. https://ojs.sgsci.org/journals/emi/article/download/243/237

Tong, L., Yan, W., & Manta, O. (2022). Artificial intelligence influences intelligent automation in tourism: A mediating role of internet of things and environmental, social, and governance investment. *Frontiers in Environmental Science*, *10*, 853302. https://www.frontiersin.org/articles/10.3389/fenvs.2022.853302/pdf

Twaha, U., Mosaddeque, A., & Rowshon, M. (2025). Accounting Implications of Using AI to Enhance Incentives for Wireless Energy Transmission in Smart Cities. https://www.allmultidisciplinaryjournal.com/uploads/archives/1744627186.pdf

Ullah, S., Nobanee, H., Azim, F., & Khiati, M. (2025). Financial foundations for sustainability: how business sophistication, tax policy, and technology shape ESG in belt and road initiative countries. Global Sustainability, 8, e3. https://www.cambridge.org/core/services/aop-cambridge-core/content/view/E7778BF8064E799EFBB5FAA5AC3F2795/S2059479825000043a.pdf/financial-foundations-for-sustainability-how-business-sophistication-tax-policy-and-technology-shape-esg-in-belt-and-road-initiative-countries.pdf

Wang, Q. (2025). Generative AI-assisted evaluation of ESG practices and information delays in ESG ratings. *Finance Research Letters*, *74*, 106757. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4998708

Wijaya, J. R. T., Prasetyo, I., Rahmatika, D. N., & Indriasih, D. (2025). ARTIFICIAL INTELLIGENCE AND AUDIT QUALITY: AN EMPIRICAL LITERATURE REVIEW FROM SCOPUS DATABASE. Fokus Ekonomi: Jurnal Ilmiah Ekonomi, 20(1), 61-76. https://ejournal.stiepena.ac.id/index.php/fe/article/download/1368/669

Wobst, J., & Lueg, R. (2025). Avoiding algorithm errors in textual analysis: A guide to selecting software, and a research agenda toward generative artificial intelligence. Journal of Business Research, 199, 115571. https://www.sciencedirect.com/science/article/pii/S0148296325003947

Xianzi, Y. A. N. G., Zheng, C., & Miao, W. A. N. G. (2025). The impact of environmental information disclosure quality on corporate value: The mediating role of innovation investment. Finance Research Letters, 79, 107304. https://www.sciencedirect.com/science/article/pii/S1544612325005677

Xu, C., Miao, Y., Xiao, Y., & Lin, C. (2025). DeepGreen: Effective LLM-Driven Green-washing Monitoring System Designed for Empirical Testing--Evidence from China. arXiv preprint arXiv:2504.07733. https://arxiv.org/pdf/2504.07733

Xu, W., Sun, Z., & Ni, H. (2024). Transparency pays: How carbon emission disclosure lowers the cost of capital. Economic Analysis and Policy, 83, 165-177. https://www.sciencedirect.com/science/article/pii/S0313592624001346

Yang, J. (2025, May). Research on ESG Audit Information Disclosure in the Coffee Industry——Taking Starbucks Coffee as an Example. In *2025 10th International Conference on Financial Innovation and Economic Development (ICFIED 2025)* (pp. 615-621). Atlantis Press. https://www.atlantis-press.com/article/126010572.pdf

Yincheng, Y., Basri, M. F., & Halim, H. A. (2024). Artificial Intelligence Enabled Corporate Governance: Enhancing Corporate Sustainability. *International Journal of Business and Technology Management*, *6*(S4), 1-22. https://myjms.mohe.gov.my/index.php/ijbtm/article/download/28340/15847/

Zhang, Y., Li, Y., Zhu, Y., & Hsu, G. C. (2023). Assessing Firm-level Climate Policy Transition Risk with AI: Evidence from China's Online Investor Interactive Platforms. Available at SSRN 5209258. https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=5209258

Zhao, J., & Gómez Fariñas, B. (2023). Artificial intelligence and sustainable decisions. European Business Organization Law Review, 24(1), 1-39. https://link.springer.com/content/pdf/10.1007/s40804-022-00262-2.pdf

Zheng, M., Li, T., & Ye, J. (2024). The Confluence of AI and Big Data Analytics in Industry 4.0: Fostering Sustainable Strategic Development. *Journal of the Knowledge Economy*, 1-37. https://link.springer.com/article/10.1007/s13132-024-02120-7

Zhu, C., Liu, X., Chen, D., & Yue, Y. (2024). Executive compensation and corporate sustainability: Evidence from ESG ratings. Heliyon, 10(12). https://www.cell.com/heliyon/pdf/S2405-8440(24)08974-6.pdf

Appendix

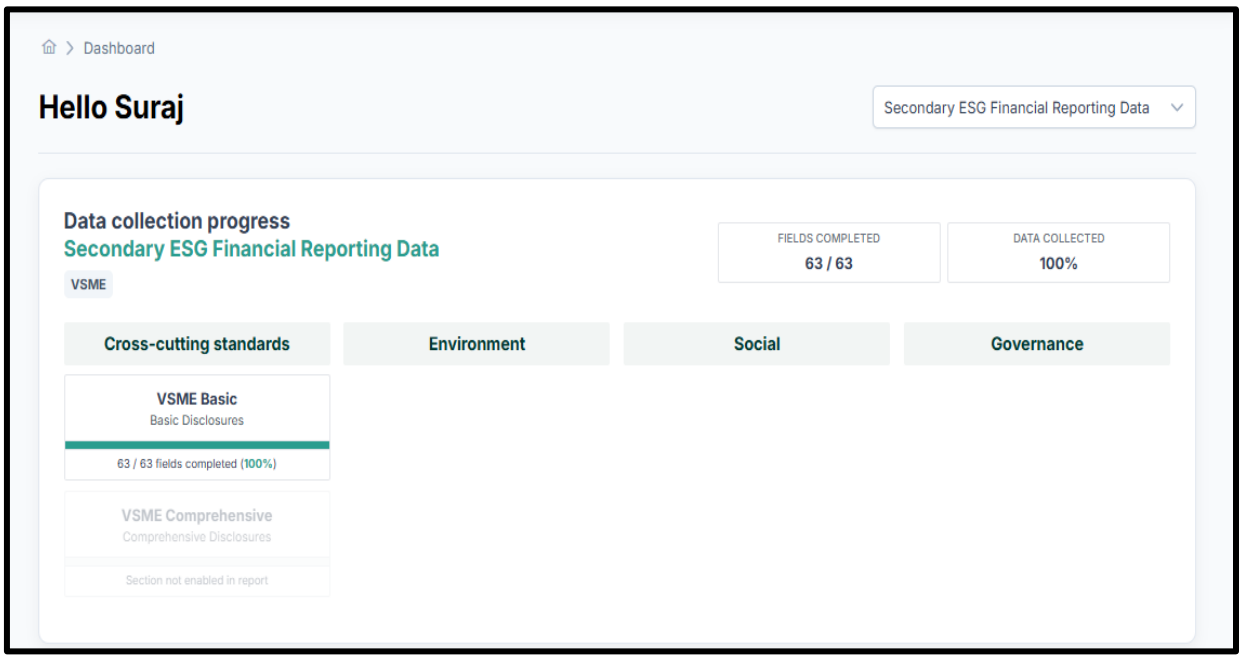


Figure 1: Dashboard of Responses

Declaration of Authenticity

I hereby declare that the present thesis entitled "AI-Assisted ESG Reporting Aligned with EFRAG VSME Standards" is my own work and that all sources and assistance used have been fully acknowledged and referenced. No part of this thesis has been submitted previously or concurrently for academic credit in any degree or diploma. I confirm that this work complies with the academic integrity and ethical standards of the institution. I am aware that any violation of these principles may result in disciplinary action, including the failure of this thesis.

Place: Berlin Germany

Date: 24-07-2025

Signature: Suraj Shrestha